

Table of Contents

Introduction	04
Introduction of Coconut	08
Recommended Coconut Varieties and Cultivation Practices	09
Establishing a Coconut Plantation	12
Planting Coconut Seedlings	14
Fertilizer Recommendations and Fertilizer Application Techniques	15
Fertilizer Recommendations for Planting	16
Fertilizer Recommendations for Young Palms	16
Fertilizer Recommendation for Adult Palms	19
Fertilizer Recommendations for King Coconut	21
Differential Fertilizer Recommendation (DFR)	23
Conservation Of Soil Moisture in Coconut Plantation	24
Moisture Conservation Methods for Young Coconut Plants	24
Moisture Conservation Methods for Mature Palms	25
Biochar for Sustainable Coconut Farming	30
Irrigation for Coconut Cultivation	33
Pests and Diseases Management in Coconut Plantations	39
Major Pests in Coconut Plantations	39
Minor Pests in Coconut Plantations	56
Diseases in Coconut Plantations and their Management	60
Weed Management in Coconut Plantation	65
Maximizing Profit Through Intercropping, Animal Husbandry, and Integrated Farming	68
Principles of Intercropping	68
Intercrops Recommended for Coconut Cultivation	69
Animal Husbandry	78
Integrated Farming in Coconut	80
Coconut Harvesting in Sri Lanka	85
Optimal Timing for Coconut Harvesting	85
Annexure	
Cash Flows of The Cropping Systems	
Annexure 1: Monocrop Coconut Cultivation	88
Annexure 2: Ginger Cultivation Under Coconut	90
Annexure 3: Turmeric Cultivation Under Coconut	91
Annexure 4: Cassava Cultivation Under Coconut	92
Annexure 5: Papaya Cultivation Under Coconut	93
Annexure 6: Black Pepper Cultivation Under Coconut	95
Annexure 7: Pineapple Cultivation Under Coconut	98
Annexure 8: Banana Cultivation Under Coconut	100
Annexure 9: Annona (Soursop) Cultivation Under Coconut	103
Annexure 10: Cashew Cultivation Under Coconut	106
Annexure 11: Mango Cultivation Under Coconut	108
Annexure 12: Cinnamon Cultivation Under Coconut	110

Target Audience

- Coconut Growers of All Scales
- Agricultural Extension Officers
- Technicians
- Researchers
- Educators
- Students
- Stakeholders in the Coconut Industry
- Policymakers

Why is this Book Important to the Target Audience?

The book is valuable for every category because it provides practical, research-based coconut-production guidance that can be applied directly to their needs: growers can improve field management and productivity, extension officers can share reliable recommendations with farmers, technicians can use it for technical support and problem-solving, researchers can treat it as a reference for further study, educators can use it to teach sound cultivation practices, students can build a strong foundation in coconut science, industry stakeholders can better understand production and value-chain requirements, and policymakers can make more informed decisions for sector development.

Introduction of Coconut

For hundreds of years, the coconut palm (*Cocos nucifera* L.) has been a great symbol of versatility. It provides food, drink, and shelter, as well as income from its products. Due to its continuous production, coconuts are readily available throughout the year or all year round. Each bunch of coconuts can contain between five to twelve nuts. At present, the world's main producers are Indonesia, the Philippines, and India.

Coconut is a source of food, oil, coconut water, coconut milk, and medicine. It has probably been used by humans for centuries. Today, the coconut is recognized as a symbol of health and wellness, reflecting its growing role in health-conscious diets and lifestyles. From the Asian tropics to South America, the coconut has spread around the globe by waves of seafaring people migrating and trading across continents. Light and buoyant, the coconut can drift vast distances on ocean currents. There are even reports of coconuts being found in the seas off Norway, far from the tropics. The purpose of this publication is to guide coconut growers and instill confidence that by adopting the recommended package of practices such as using high-yielding coconut varieties, applying regular fertilizers and organic manure, conserving soil and moisture, providing irrigation, and managing plant health, average coconut yields or mean coconut yields can be increased from the present level of 45-55 nuts to 80-100 nuts per palm per year.

Recommended Coconut Varieties and Cultivation Practices

Selection of Coconut Cultivars for Establishing a Coconut Plantation

The coconut palm is a perennial plantation crop that occupies the land for several decades. Therefore, when starting a new plantation, it is important to select the most suitable variety considering the soil, climate, and available space for cultivation, and to use good-quality planting material.

Besides, the planting material selection, proper planting and aftercare should be followed. This includes maintaining ideal plant density, adopting appropriate spacing for mono/intercropping, and following other recommended aftercare practices.

Coconut Research Institute has recommended the following coconut varieties for cultivation.

CRIC 60 (Tall × Tall)

- This variety is a selection of tall palms for superior traits.
- Recommended for cultivation in all coconut-growing areas and for large-scale plantations.
- Begins bearing coconuts in 5-6 years under normal management conditions.
- The yield potential is about 12,000-15,000 nuts per hectare per year.
- Seedlings are certified using a dark blue colour label.
- Available at the Coconut Research Institute, Coconut Cultivation Board and private registered nurseries



CRISL 98 (Tall × San Ramon Variety)

- A hybrid between the Sri Lankan Tall variety and the San Ramon Tall variety of the Philippines.
- Ideal for wet and intermediate zones, and for large-scale plantations.
- Standardised to coconuts in 5-6 years under normal management conditions.
- Yield potential is approximately 12,000-15,000 nuts per hectare per year.
- Produces higher copra yield due to increased nut size and kernel thickness.
- Seedlings are certified using a dark red colour label.



CRIC 65 (Dwarf × Tall)

- A hybrid between the Sri Lankan Dwarf variety and Sri Lankan Tall variety.
- The yield potential is about 20,000-22,000 nuts per hectare per year.
- Standardised to coconuts in 3-4 years under normal management conditions.
- Seedlings are certified using a dark green colour label.



Kapruwana (CRISL 2004 / DG × SR)

- A hybrid between the Sri Lankan Green Dwarf variety and the San Ramon Tall variety from the Philippines.
- The yield potential is about 20,000-22,000 nuts per hectare per year.
- Begins bearing coconut in 3-4 years under normal management conditions.
- Seedlings are certified using a dark orange colour label.



Kapsuwaya (CRISL 2012 / DB × T)

- A hybrid between the Sri Lankan Brown Dwarf variety and the Sri Lanka Tall variety.
- The yield potential is about 20,000-22,000 nuts per hectare per year.
- Begins bearing coconuts in 3-4 years under normal management conditions.
- Seedlings are certified using a pink colour label.



Kapsetha (CRISL 2013 / DB × SR)

- A hybrid between the Sri Lankan Brown Dwarf variety and the San Ramon Tall variety from the Philippines.
- The yield potential is about 20,000-22,000 nuts per hectare per year.
- Begins bearing coconut in 3-4 years under normal management conditions.
- Seedlings are certified using a black colour label.



CRISL 2020 (Sri Lankan Tall x Malayan Red Dwarf)

- A hybrid between the Sri Lankan Tall variety and the Malayan Red Dwarf variety.
- The yield potential is about 22,000 nuts per hectare per year.
- Begins bearing coconut in 3-4 years under normal management conditions.
- Seedlings are certified using a purple colour label.



In general, the hybrid cultivars are recommended for cultivation in home gardens as well as in large-scale plantations for wet and intermediate zones. They may also be grown in other areas in Sri Lanka with advanced crop management conditions, with a permanent water supply for irrigation.

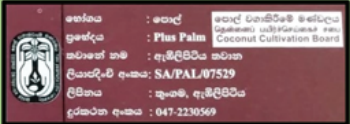
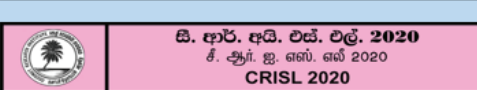
Plus Palm

- Seed coconuts are collected from selected mother palms of the tall variety with high yield potential.
- Yield potential is approximately 10,000-12,000 nuts per hectare per year.
- Begins bearing coconuts in 5-6 years under normal management conditions.
- Available in the nurseries of the Coconut Cultivation Board.

Moorock Tall

- A variety selected from the Moorock Estate located in Mawathagama.
- Recommended for cultivation in wet and intermediate zones of Sri Lanka.
- Yield potential is approximately 11,000 nuts per hectare per year.
- Begin bearing coconut in 5-6 years under normal management conditions.
- Seedlings are certified using a white colour label.
- Available in the nurseries of the Coconut Cultivation Board.

Coconut Cultivars Suitable for Different Climatic Zones

CRIC 60 / Plus Palm These tall varieties are recommended for all coconut growing areas in Sri Lanka	 සී. ආර්. අයි. සී 60 சீ. ஆர். இ. சீ 60 CRIC 60 
CRISL 98 / Moorock Tall These cultivars are recommended for large-scale cultivation in wet and intermediate zones of Sri Lanka	 සී. ආර්. අයි. එස්. එල්. 98 சீ. ஆர். இ. எஸ். எல் 98 CRISL 98 
CRIC 65 / Kapruwana / Kapsetha / Kapsuwaya / CRISL 2020 These hybrids are recommended for home gardens and for large-scale cultivation in wet and intermediate zones. They are also recommended for the rest of the areas in Sri Lanka under advanced crop management conditions with a permanent water supply for irrigation.	 සී. ආර්. අයි. සී 65 சீ. ஆர். இ. சீ 65 CRIC 65    
Characteristics of a high-quality coconut seedling • Possesses a large, strong stem / collar region • Has broad, dark green leaves • Convex leaf surface • Sharp ridges of leaves • Short petioles connecting the fronds to the stem • Free from pest and disease damage	

Establishing a Coconut Plantation

Seedling density at coconut planting

When planting coconut as a mono-crop, it is recommended to maintain a plant spacing of 8 m x 8 m (26 feet x 26 feet) (158 plants per hectare).

1. Square method

Space between seedlings		No. of plants	
Meters	Feet	Per hectare	Per acre
8.0 × 8.0	26 × 26	158	64

Coconut plantations with the above spacing are recommended for intercropping with annual crops or with perennial crops that are less competitive.

2. Wide row method

Space between seedlings		No. of plants	
Meters	Feet	Per hectare	Per acre
8.0 × 10.0	26 × 32	128	52

3. Special wide row method

Space between seedlings		No. of plants	
Meters	Feet	Per hectare	Per acre
8.0 × 12.0	26 × 40	102	42

The above spacing is recommended for coconut cultivation when tea or sugar cane is intercropped to generate additional income.

Underplanting and Replanting

Underplanting or Replanting should be carried out when;

- The majority of coconut palms in a coconut plantation are over 60 years old, and/or the plantation does not provide economic yield

- Yield has reduced to approximately 1,000 nuts per acre per year, and
- Palms have grown very tall, making harvesting and other cultural operations difficult, and the number of palms per acre has decreased to 50.

1. Underplanting

Underplanting means planting seedlings between the rows of old palms and gradually removing old palms over a period of 5-6 years. First, the new planting holes should be marked according to the selected planting design inbetween the rows of the old palms. Then, very poor-yielding and those close to the identified planting pits (within 8 ft circle) should be removed and the new seedlings should be planted as per the recommendations of the Coconut Research Institute.



Underplanting in an existing coconut plantation

2. Replanting

Replanting is the complete removal of an existing plantation and the establishment of a new plantation.



Replanting after the complete removal of the old palm

According to the research findings, precocity, growth, and nut production of palms in the replanted cultivation are higher than in the underplanting system.

Therefore, replanting methods are recommended for advanced management and large-scale coconut plantations. Underplanting is recommended for small-scale growers to avoid economic hardships in replanting.

3. Vacancy Filling

To keep the plantation uniform, after one year of planting, all weak coconut plants should be removed and replaced with healthy seedlings. This process should be completed within a period of 3-5 years from the first day of planting. It is not advisable to fill the vacancies after that. But if a high number of vacancies and sufficient spaces are available, it is recommended to fill the vacancies.

Planting Coconut Seedlings

- a. The size of the planting hole should be determined according to the soil type. The length, width, and depth of the planting hole should be 0.9 m × 0.9 m × 0.9 m for sandy loamy soil, and 1.2 m × 1.2 m × 1.2 m for gravel soil.
- b. When planting hole is cut, place the topsoil up to a depth of about 30 cm to one side, and the remaining subsoil separately on the other side of the surface of the planting hole.
- c. Place coconut husks at the bottom of the pit, the first layer with the coir side upwards, and cover using topsoil. Later, place the second layer of husks and cover with the remaining topsoil. About 30-35 coconut husks are required for placing two layers in a planting hole. Each husk can absorb 6-7 times its weight in water. During dry periods, this water is gradually released into the soil, providing moisture for the roots of young palms. Prepare a mixture using organic manure (10 kg of dry cow dung or 5 kg of goat manure), 1 kg of dolomite and young palm mixture (YPM), depending on the climatic zone or straight fertilizer given in Tables 1 or 2, along with subsoil, and subsoil to fill the rest of the planting hole.



Planting coconut seedlings and other care



Fertilizer Recommendations and Fertilizer Application Techniques

Nutrients are lost from the soil annually when nuts, fronds, spathes, and other parts of the palm are harvested for human use. Additional nutrient losses occur through weed absorption, soil erosion, and leaching. Due to the reduction of essential nutrients in the soil, the yield gradually declines. To achieve sustainable yields, essential nutrients (nitrogen, phosphorus, potassium, and magnesium) should be applied continuously from planting through the seedling and mature stages. Provide nitrogen (N) to coconut cultivation, urea (a chemical fertilizer), or organic manures such as cow dung, goat manure, seasoned poultry manure, or green manures such as *Gliricidia*, and compost can be used.

- Provide Phosphorus (P), use Eppawala Rock Phosphate (ERP) for wet and intermediate zones, and Triple Super Phosphate (TSP) for dry zone.
- Although Potassium (K) is contained in coconut husks, wild sunflowers, banana stems, and animal manure to some extent, it is not sufficient for the growth and proper yielding of a coconut palm. Apply a potassium fertilizer (Muriate of Potash-MOP) to provide the K nutrient.
- Dolomite is recommended as Magnesium (Mg) nutrient.

The nutrient requirements of a coconut palm vary with its growth stage. Young (juvenile plants before bearing) require high nitrogen, and phosphorus for a vegetative growth. An adult palm (bearing palms) requires potassium and magnesium for generating high yields.

In wet and intermediate zones, use Eppawala Rock Phosphate as the phosphorus source; in the dry zone, use Triple Super Phosphate as the phosphorus source.

The nutrient requirements of a young plant follow the order (high to low)

N > P > K > Mg

The nutrient requirements of an adult palm follow the order (high to low)

K > Mg > N > P

Accordingly, separate chemical and mineral (inorganic) fertilizers, organic manures and mixtures of organic and chemical fertilizers have been recommended by the Coconut Research Institute at the time of planting, young, and adult plantations. Chemical and mineral fertilizers can be applied in the form of straight fertilizer or mixtures of fertilizer.

Fertilizer Recommendations for Planting

Fertilizer mixture requirements at the time of planting

Wet and Intermediate Zone		Dry Zone	
Fertilizer Type Per Planting Hole	Quantity Per Planting Hole	Fertilizer Type	Per Planting Hole
Young Palm Mixture – W (YPM – W)	1250 g	Young Palm Mixture – D (YPM – D)	850 g
Dolomite	1000 g	Dolomite	1000 g
Goat or cattle manure	10 kg	Goat or cattle manure	10 kg

Composition of YPM-W (%)

$N - P_2O_5 - K_2O \longrightarrow 11-16-14$

Composition of YPM-D (%)

$N - P_2O_5 - K_2O \longrightarrow 15-15-20$

Straight fertilizer requirements at planting

Wet and Intermediate Zone		Dry Zone	
Fertilizer Type	Quantity for a Per Planting Hole	Fertilizer Type	Quantity for a Per Planting Hole
Urea	250 g	Urea	250 g
Eppawala Rock Phosphate	750 g	Triple Super Phosphate	350 g
Muriate of Potash	250 g	Muriate of Potash	250 g
Goat or cattle Manure	10 kg	Goat or cattle Manure	10 kg

When only organic manures are applied at the time of planting, subsoil should be mixed with 20 kg of cow dung or 15 kg of goat manure, 1 kg of Eppawala rock phosphate, and 1 kg of dolomite.

Fertilizer Recommendations for Young Palms

The following fertilizer mixtures are recommended for young coconut palms according to their age.

Inorganic Fertilizer

i. Straight fertilizer

a) For wet and intermediate zones

Age of the coconut plant	6 Months	1 Year	1 ½ Years	2 Years	2 ½ Years	3 Years	3 ½ Years	4 Years up to bearing (once every 6 months)
Type of Fertilizer	Amount (g/palm)							
Urea	190	235	235	305	305	375	375	470
Eppawala Rock Phosphate	420	530	530	690	690	850	850	1060
Muriate of Potash	190	235	235	305	305	375	375	470
Dolomite	500	500	500	500	500	500	500	500

b) For dry zone

Age of the coconut plant	6 Months	1 Year	1 ½ Years	2 Years	2 ½ Years	3 Years	3 ½ Years	4 Years up to bearing (once every 6 months)
Type of Fertilizer	Amount (g/palm)							
Urea	190	235	235	305	305	375	375	470
Triple Super Phosphate	160	200	200	300	300	360	360	400
Muriate of Potash	190	235	235	305	305	375	375	470
Dolomite	500	500	500	500	500	500	500	500

ii. Fertilizer mixtures

a) Young palm mixture

Age of the coconut plant	Rate of application (g/palm)		
	YPM-W (for wet and intermediate zones)	YPM-D (for dry zone)	Dolomite
6 Months	800	540	500
12 Months (1 Year)	1000	670	500
18 Months (1 ½ Years)	1000	670	500
24 Months (2 Years)	1300	905	500
30 Months (2 ½ Years)	1300	905	500
36 Months (3 Years)	1600	1110	500
42 Months (3 ½ Years)	1600	1110	500
4 Years up to bearing	2000	1340	500

Organic and Inorganic Fertilizer Mixtures

1. Other fertilizers for young coconut palms with cow dung

Nutrient	Source of Fertilizer	Age of the coconut plant				
		6 Months	1 Year	2 Years	3 Years	4 Years up to bearing
N	Cattle manure (kg) (moisture 20% - 30%)	5	12	16	20	25
P	Eppawala Rock Phosphate (ERP) (g)	200	450	600	750	1000
Mg	Dolomite (g)	250	250	250	250	250
K	Muriate of Potash (MOP) (g) or	30	200	250	325	400
	Natural Potassium Sulfate (K ₂ SO ₄) (g)	50	240	300	390	480

2. Other fertilizers for young coconut palms with goat manure

Nutrient	Source of Fertilizer	Age of the coconut plant				
		6 Months	1 Year	2 Years	3 Years	4 Years up to bearing
N	Cattle manure (kg) (moisture 20% - 30%)	3	7	9	11	13
P	Eppawala Rock Phosphate (ERP) (g)	220	500	670	850	1150
Mg	Dolomite (g)	500	500	500	500	500
K	Muriate of Potash (MOP) (g) or	50	130	170	230	300
	Natural Potassium Sulfate (K_2SO_4) (g)	60	150	200	270	370

3. Other fertilizers for young coconut palms with seasoned poultry manure

Nutrient	Source of Fertilizer	Age of the coconut plant				
		6 Months	1 Year	2 Years	3 Years	4 Years up to bearing
N	Seasoned poultry manure (kg) (moisture 20% - 30%)	5	12	15	20	23
P	Eppawala Rock Phosphate (ERP) (g)	100	225	300	400	600
Mg	Dolomite (g)	250	250	250	250	250
K	Muriate of Potash (MOP) (g) or	50	125	150	190	240
	Natural Potassium Sulfate (K_2SO_4) (g)	60	150	180	230	290

4. Other fertilizers for young coconut palms with Gliricidia

Nutrient	Source of Fertilizer	Age of the coconut plant				
		6 Months	1 Year	2 Years	3 Years	4 Years up to bearing
N	Gliricidia (kg) (moisture 60% - 70%)	5	12	16	20	25
P	Eppawala Rock Phosphate (ERP) (g)	275	630	830	1040	1375
Mg	Dolomite (g)	250	250	250	250	250
K	Muriate of Potash (MOP) (g) or	60	150	200	250	270
	Natural Potassium Sulfate (K_2SO_4) (g)	75	180	250	300	375

5. Other fertilizers for young coconut palms with agricultural compost (N:P:K – 1:0.5:1)

Nutrient	Source of Fertilizer	Age of the coconut plant				
		6 Months	1 Year	2 Years	3 Years	4 Years up to bearing
N	Agricultural compost (kg)* (moisture 25%)	8	19	25	31	39
P	Eppawala Rock Phosphate (ERP) (g)	200	460	600	770	1000
Mg	Dolomite (g)	250	250	250	250	250
K	Muriate of Potash (MOP) (g) or	-	100	125	125	200
	Natural Potassium Sulfate (K_2SO_4) (g)	-	130	150	150	250

Fertilizer Recommendation for Adult Palms

The amount of fertilizers that should be applied to an adult coconut palm per year is mentioned below.

Inorganic fertilizer

i. Straight fertilizer

Type of Fertilizer	Amount (g/palm)	
	Wet and Intermediate Zone	Dry Zone
Urea	800	800
Eppawala Rock Phosphate (ERP)	900	-
Triple Super Phosphate	-	400
Muriate of Potash (MOP)	1600	1600
Dolomite	1000	1000

ii. Fertilizer mixtures

Amount to be applied (kg/palm/year) from Adult Palm Mixture (APM)		
APM-W (for wet and intermediate zones)	APM-D (for dry zone)	Dolomite
3.3	2.8	1

Composition of APM-W (%)

$N - P_2O_5 - K_2O \longrightarrow 11-8-29$

Composition of YPM-D (%)

$N - P_2O_5 - K_2O \longrightarrow 13-7-34$

It is recommended to apply 1 ½ times from the above fertilizer rates for the improved varieties such as CRIC 60, CRIC 65, CRISL 98, Kapruwana, Kapsuwaya, Kapsetha, and high-yielding palms that produce more than 75 nuts/palm/year.

Recommended fertilizer packages for adult coconut palms with organic manures

Organic manures and other sources		Amount (kg)	Muriate of Potash (g)	Natural Potassium Sulfate (g)
1	Cattle manure (Moisture 20% - 30%)	30	1250	1500
2	Goat manure (Moisture 20% - 30%)	25	800	1250
3	Seasoned poultry manure (Moisture 20% - 30%)	30	750	1000
4	Gliricidia (Moisture 60% - 70%)	30	1000	1350
	Eppawala Rock Phosphate (ERP)	0.7		
	Dolomite	0.5		

- Poultry manure removed from poultry farms should be applied to the coconut palms only after seasoning them in the field for 2-3 months.
- Depending on the availability of organic manures, any of the above five mixtures can be used for adult coconut palms.
- For organic farming, natural Potassium Sulfate should be used instead of Muriate of Potash.

Application of organic manure and inorganic fertilizer combinations to adult coconut palms

Organic Manure	Urea (g/plant)	(i) ERP or (ii) TSP (g/plant)	MOP (g/plant)	Dolomite (g/plant)
Cattle manure (15 kg)	400	i) - 450	1400	500
(Moisture 20% - 30%)		ii) - 200		
Goat manure (13 kg)	400	i) - 450	1200	500
(Moisture 20% - 30%)		ii) - 200		
Seasoned poultry manure (15 kg)	400	i) - 450	1175	500
(Moisture 20% - 30%)		ii) - 200		
Gliricidia (15 kg)	400	i) - 700	1300	750
(Moisture 60% - 70%)		ii) - 350		

It is highly recommended to use a combination of organic manures and chemical fertilizers for long-lasting effects and good retention of soil nutrition.

Fertilizer Recommendations for King Coconut

The same fertilizer recommendations given above can be used for King Coconut at planting, for young palms and adult bearing palms.

Straight fertilizer recommendation for toddy tapping palms

Type of Fertilizer	Amount (g/palm /year)	
	Wet and Intermediate Zones	Dry Zone
Urea	1500	1500
Eppawala Rock Phosphate	2100	-
Triple Super Phosphate	-	950
Muriate of Potash	4800	4800
Dolomite	1000	1000

Method of Fertilizer Application

1. Fertilizer application for young palms

- Depending on the age of the young palm, broadcast the recommended amount of fertilizers around the palm basin to the distance shown below.
Example: Spread the recommended amount of fertilizer around the palm within 60 cm from the base for a one-year-old palm.
- Cover the area where the soil is turned and fertilized to a depth of about 10-15 cm with any material such as weed slash, coconut fronds, or coconut husks.

Distance from the base of the coconut palm for application of fertilizer

Age of Palm	Distance from the palm base (cm)
6 Months	30
1 Year	60
2 Years	90
3 Years	120
From 4 Years up to bearing	150
For an adult palm	180

2. Fertilizer Application for Adult (Bearing) Palms

- The soil should be moistened when applying fertilizer.
- Fertilizer should be applied annually to bearing palms. For sandy soils, it is preferable to apply the recommended dosage of fertilizers every 6 months as split applications, rather than applying the entire quantity at once.

i) Application of fertilizers to adult coconut palms can be done in two ways: surface application or trench application.

Surface application

In an adult coconut palm, "area within a 1.8 m (6 ft) radius from the base constitutes the manure circle. Weeds within this area should be slashed, avoiding complete removal by hoeing. Remove the mulching materials such as coconut fronds and coconut husks, if present, from the manure circle without removing the topsoil. Broadcast the recommended quantity of fertilizers in the manure circle, leaving a distance of 15 cm from the palm. Mix fertilizers and soil well by digging with a hoe or fork about 10-15 cm deep. After digging or hoeing, cover the fertilized area with mulch.



Spreading of fertilizer
in the manure circle



Mixing fertilizer with soil



Mulching of manure circle

- Prepare 90 cm wide and 15 cm deep trench which is 90 cm away from the base around the palm and apply the recommended quantity of fertilizer into the trench. Mix applied fertilizer with soil and cover with the soil that was removed during preparation of the trench. Finally, the manure circle (which is an area of 1.8 m radius circle from the base of the palm) should be covered with any mulching material such as coconut fronds, coconut husks, leaving 15 cm radius from the base of the palm.
- Since there is excessive surface water flow in plantations with steeper slopes, fertilizers should be applied by establishing semicircle trenches with a depth of about 10 cm on the upper side of the slope.
- The radius of the inner circle of the trench should be 0.5 m and the radius of the outer circle should be about 1.5 m. After applying fertilizer inside the trench, it should be mixed with the soil in the trench, covered well with soil and then with mulch.

Differential Fertilizer Recommendation (DFR)

In this method, fertilizers are recommended for a particular site, based on the analysis of nutrient content of the 14th frond leaves. As the nutrient levels of the palm are compared with the sufficiency nutrient ranges of coconut palm, this provides accurate information about the extent of soil nutrient deficiency and any hidden nutrient deficiencies. This enables correction of nutrient deficiency of palms at an early stage. Additionally, this also prevents fertilizer wastage and gives the most appropriate fertilizer recommendation for the site.

Selection of palms by observing the field and leaf sampling are the two main activities in the field. Before palm selection, it is important to observe the field carefully and identify fields with same conditions. Field should be blocked according to differences of characteristics such as age of the palms, cultivar, soil type, shape of the land, presence of intercrops, animal husbandry, economical value of the plantation (if it is successfully underplanted), irrigation, and nutrient deficiencies. Then, palms are selected for leaf sampling in separate blocks. Leaf samples can be taken from at least three palms per selected field block. It is important to have the following characteristics in the selected palms for leaf sampling.

- A complete manure circle
- A palm without mutations
- Having a manure circle without termite mounds
- Without having much forested surroundings
- No weeds in the manure circle
- No insect attacks on the leaves
- A palm yields approximately 50 nuts per year

For selection of the 14th leaf, leaves are counted spirally considering fully opened leaf as the 1st leaf. In addition, following characters could also be considered for easy identification of the 14th frond.

1. It is located parallel to the ground
2. It has fist- sized nuts (size can be slightly changed with cultivar)
3. It is immediately below the flower which is next to open

Note: Leaf sampling should not be carried out in severe drought or rainy conditions.

After selecting the 14th leaf, six leaflets should be removed from the middle area of the mid-rib of the frond, as three leaflets from each side of the mid rib, using a stainless-steel knife. After that leaflets are kept as bundle and 20 cm size part is cut from the middle area.

Keep the cut leaflets in dry paper bags with labels on sampling date, name of the estate, block number, sample number, sampled frond number prior to bringing them to the laboratory, for analysis. Printed bags should not be used to keep the samples, as they may contaminate the samples with unnecessary substances.



14th frond



Selection of appropriate leaflets from the frond



Cutting the leaflets after removing the ekels

Conservation of Soil Moisture in Coconut Plantations

Continuous water supply is essential for a coconut palm to achieve sustainable nut production throughout the year. A young coconut palm requires 10–30 L of water per day, while an adult (bearing) palm requires around 40–60 L per day.

Large number of roots of coconut palm are extended from the soil surface to a depth of one meter and in a circular area with a radius of 2 meters. 2–5 mm long water absorption areas are located at the end of each root. If a dry climate persists for a long time without sufficient moisture, the water-absorbing areas of the roots become covered with a thick protective layer that cannot absorb water, and thereafter the roots lose their ability to absorb water.

Therefore, conserving soil moisture by retaining rainfall and reducing evaporation is essential.

Moisture Conservation Methods for Young Coconut Plants

Coconut palms can suffer significant damage if they are subjected to drought at all stages of their growth. In order to prevent this, the following practices should be adopted:

1. Application of Mulch

- Manure circles in the young plantation should be mulched with coconut fronds, husks, or similar organic material.

2. Burial of Coconut Husks

- This should also be done continuously for 2–5 years after planting.
- For two-year-old plants, construct semi-circular trenches 60 cm wide and 60 cm to 75 cm depth, leaving 90 cm from the plants, and bury the coconut husks.

The distance from the base of the coconut plants to the trench varies depending on their age as given below.

Age of Palm	Distance from Young Palm to the Trench
2 Years	0.9 m (3 ft)
3 Years	1.2 m (4 ft)
4 Years	1.5 m (5 ft)
5 Years	1.8 m (6 ft)

3. Control of cover crops / grasses / weeds

- During the drought period, cover crops, grasses and other weeds will compete with coconut palms for soil moisture.
- In such situations, the damage to the plantations is relatively high.
- Cover crops, grasses and other weeds in a coconut plantation should be properly managed by an integrated approach such as regular grazing with animals or slashing.

Moisture Conservation Methods for Mature Palms

1. Establishment of Drains

- i. Contour Drains
- ii. Drainage Drains

i. Contour Drains

- The drains that are established perpendicular to the slope of the coconut land are called contour drains.
- Contour drains are recommended only when the slope of coconut land is more than 3%.
- Spacing to be adopted between contour drains is as follows:

Slope	Intensity of Slope	Distance between Drains
5%	1:20	40 m (133 ft)
10%	1:10	20 m (66 ft)
15%	1:7	15 m (50 ft)
20%	1:5	10 m (33 ft)

- Soil clumps should be left in the drain at a distance of 35 m to 50 m to reduce the speed of water flowing in the drain.



Contour drains established in coconut plantation

ii. Drainage Drains

- Due to the long-term inundation of low land areas of coconut plantations, the growth of young palms and productivity of adult palms decrease.
- Establishing drainage drains between the rows and connecting them allows excess water to be efficiently drained from the land



Drainage drains established in coconut plantations

2. Burial of Coconut Husks

- Coconut husks have a special ability to absorb water up to 6-7 times of their weight.
- Use fresh coconut husks for burial.
- Buried coconut husks absorb water during the rainy season and release it slowly for the palms during the dry. The damage to coconut palms due to drought can be reduced by this method.
- The pits should be filled up to the level of the soil surface by placing layers of coconut husks and soil alternately, and the remaining soil should be heaped sufficiently to fully cover the coconut husks, preventing black beetle breeding.
- Husks buried once can give successful results for 5-6 years.

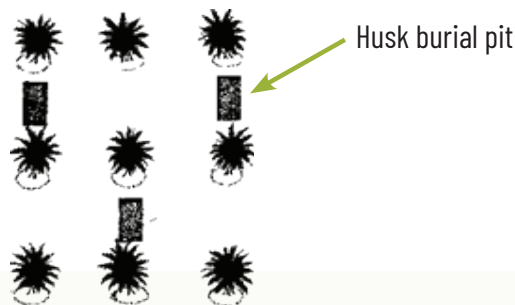


Cross section of a pit applied with coconut husk

Methods of Husk Burial

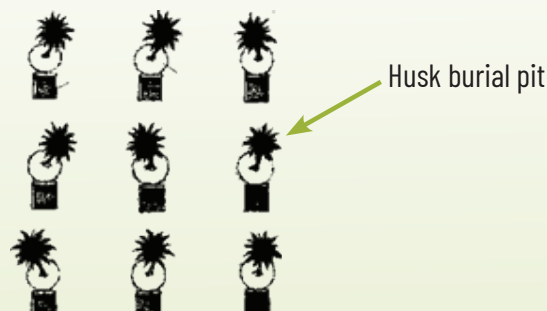
i). One husk pit per two palms in the coconut row

- Establish pits (alternatively) 2.4 m (length) × 1.2 m (width) × 0.9 m (depth) between two palms in a coconut row.
- This pit can accommodate more husks than individual pits due to its larger volume.
- One pit requires about 350 – 400 coconut husks.
- This method is most suitable for husk burial.



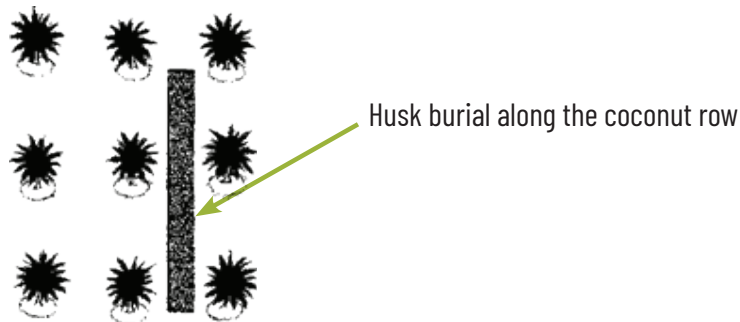
ii). Individual husk pits for each palm

- Prepare a single coconut pit of 1.2 m (length) × 1.2 m (width) × 0.9 m (depth) for each palm.
- One pit requires about 150 – 200 coconut husks.



iii). Burying coconut husks as a drain along the coconut row

- Coconut husks can be buried by establishing 0.9 m (width) x 0.9 m (depth) long drains along the coconut row, 2.0 m away from the coconut palm.
- Suitable for lands with a surplus of coconut husks and high coconut yields. Drains should be positioned perpendicular to the slope.



3. Burial of Coir Dust

- The preparation of coir dust burial pits is similar to that of coconut husk burial pits.
- Alternate layers of coir dust and soil are to be used to fill the pit.
- The coir dust pit should be covered with a layer of soil and raised from the ground level to prevent the breeding of black beetles.
- The length, width and depth of this pit should be 2.4 x 1.2 x 0.9 m, respectively.

4. Covering Manure Circle / Mulching

- Mulch is to be applied covering the manure circle area, leaving a 15 cm gap from the base of the coconut palm.
- Mulching:
 - Reduces soil erosion.
 - Conserves moisture in the manure circle.
 - Reduces soil temperature.

Common materials used as mulch

- Coconut husks
- Coconut fronds
- Pruned leaves, branches, and slashed weeds
- Paddy straw

i. Use of coconut husks as a mulch



Use of coconut husk for mulching

- Apply only two layers of coconut husks as mulch.
- Place the coir side of the coconut husk facing down for best results. Coconut husks should be applied, leaving 6 inches (15 cm) from the base of the palm.
- A coconut palm requires around 200-250 coconut husks.

ii. Use of coconut fronds as a mulch



Use of coconut fronds for mulching

- Cut the fallen coconut fronds into 2-3 pieces and cover the manure circle well.
- The base of the frond (petiole) can be used as fuel wood.
- Coconut fronds decompose more rapidly than coconut husks.

iii. Use of pruned leaves, branches, and slashed weeds



Applying pruned gliricidia leaves as mulch

- Because gliricidia leaves decompose quickly, they are used as green manure rather than mulch. Acacia leaves can also be used as mulch.
- Branches of other trees, slashed weeds, grass, aquatic weeds such as salvinia, water hyacinth can also be used as mulch.

iv. Use of paddy straw as a mulch

- The left-over paddy straw during the harvesting season can be used as mulch.
- It is not suitable to use this as mulch for young coconut plants, as termites can breed easily.

5. Application of Organic Manures

- Applying organic manures instead of chemical fertilizers increases the amount of humus in the soil.
- This improves soil moisture and nutrient holding capacity, aeration, structure and microbial activity.

Ex: Goat manure, cow dung, poultry manure and compost

6. Preparation of Contour Fences

- Establish contour fences along the same elevations against the slope of the land.
- Stones in the land can be used for this purpose.
- By widening the top of the fence about 60 cm, its stability can be increased.

7. Use of Different Cropping Methods

- Because of the wide spacing between coconut palms, the soil is prone to erosion in the rainy season and drying in summer.
- Soil moisture can be conserved by growing one or more crops to cover open spaces and slow water flow.
Ex: Intercropping, mixed cropping, and planting of coconut and other crops according to the SALT method.

8. Establishment of Live Fences (SALT Method)

- This method is used for lands where drains alone are not sufficient to prevent soil erosion.



Established SALT method

- This allows sloping lands to be safely used for growing crops

Establishment of live fences and their maintenance:

1. Establish stone fences at required places.
2. Plant gliricidia and other nitrogen-fixing trees in two contour rows with 30 cm separating each plant.
3. Cultivate inter-crops with coconut in the fields where gliricidia fences are established (E.g.: Cocoa, Rambutan, Black Pepper)
4. Cultivate inter-crops in crop rotation.
5. Obtain green manure for the coconut plantation by promptly pruning gliricidia branches.
6. Strengthen the area of the gliricidia fence using twigs, stones, and other materials.

9. Loosening the Soil (Compacted by the Animal's Hooves and Heavy Machinery Usage)

- As irregular cattle management causes soil compaction, the reduction of pore spaces in the soil reduces water absorption into the soil.
- Fertile soil on the surface is washed away as water flows from the soil surface increases.
- In such lands, loosening the soil before the rainy season increases the absorption of water into the soil.

Important Guidelines for Preventing Drought Damage to Coconut Plantations

1. Preserve the water resources on the land (by blocking the drains)
2. Utilize irrigation for maximum efficiency
3. Avoid removal of mulch
4. Avoid planting of coconut seedlings during drought
5. Avoid the application of chemical fertilizers during drought
6. Avoid harrowing the land during drought
7. Reduce the growth of cover crops and weeds
8. Prevent incidence of fire in the plantation
9. Manage free grazing of animals in the coconut land
10. Mulch the manure circle whenever possible by using dry coconut fronds
11. Tying coconut fronds during drought and releasing them during rains
12. Provide temporary shade
13. Planting drought resistant coconut varieties

Biochar for Sustainable Coconut Farming

Biochar is a high-carbon substance produced by slow pyrolysis, where heating is generated under low or no oxygen conditions biochar application is considered a lifetime investment for coconut plantations due to its long-lasting benefits in soil improvement.



Unlike traditional fertilizers that need frequent reapplication, biochar remains stable in the soil for decades or even centuries, acting as a persistent soil amendment.

It improves soil fertility by,

- ✓ Increasing water holding capacity of the soil
- ✓ Promoting microbial activity in the soil,
- ✓ Enhancing the soil's nutrient retention capacity, and
- ✓ Increasing other soil properties in a beneficial way

Biochar can be produced from a diverse range of biomass feedstocks, primarily organic materials rich in carbon. Common raw materials include Gliricidia stems, king coconut husks, coconut fronds, paddy husks, Guinea grass, other weed residues, wood shavings, wood dust, and coconut spathes. These materials are readily available and serve as sustainable options for biochar production.

Meeting the growing demand for high-quality biochar to support coconut cultivation has posed technological challenges in achieving mass production efficiency. The Coconut Research Institute of Sri Lanka, has modified a double chamber pyrolyzer to introduce as a commercially viable pyrolysis system enabled by auto-combustion with minimal external energy inputs.

The designed double-chamber mini pyrolyzer produces biochar by slow pyrolysis at 450-600 °C. Structurally, it contains an external combustion chamber encasing an inner insulated pyrolysis chamber, connected by a vertical chimney. Self-sustained operation ignites initial combustion at the top, with the reaction front propagating downwards as required air enters from the bottom inlet holes. The arrangement separates pyrolysis from heating, while the insulating layer preserves inner temperatures.



a) Outer chamber

b) Inner chamber

c) Chimney

Dimensions of the double-chamber pyrolyzer: (a) Outer chamber, (b) Inner chamber, and (c) Chimney

This innovative configuration aims to balance product quality, production scalability, and automation needs for a commercially feasible biochar production system designed for coconut farming applications. The use of coconut residue streamlines supply logistics while the efficient pyrolyzer maximizes yield, minimizing external energy demands through internal combustion. Ongoing enhancements to modularity, combustion control, and byproduct capture can help realize a robust, sustainable pyrolysis platform for decentralized biochar generation.

To produce biochar with this unit, cut the raw materials into small, uniform pieces to ensure even carbonization. These pieces should then be spread out in a sunny, dry area and sun-dried until they reach a moisture content of approximately 30–35%. Sufficiently dried feedstock is necessary to prevent incomplete pyrolysis. Once adequately dried, place the raw materials loosely into the inner chamber of the pyrolysis unit and seal it with a lid to create a limited oxygen environment. Then position the inner chamber inside the outer chamber and fill the space between the two chambers with combustible material such as firewood or dried biomass. Ignite this firing material from the top of the outer chamber to allow for uniform downward burning. Once proper firing is observed, seal the unit with a chimney to regulate airflow and maintain combustion efficiency. After 1 to 2 hours of burning, high-quality biochar can be obtained, ready for cooling and later use.

Biochar Recommendations for Coconut Plantations

For planting coconut seedlings, 30 kg of biochar with 20% moisture content should be incorporated into the soil of the planting hole. This improves soil quality and nutrient availability, supporting better establishment and growth.

Biochar application for planting Coconut seedlings



For adult coconut palms, biochar can be applied either as a one-time application or through multiple applications. In the one-time application method, maintain a distance of 1.0 m from the base of the palm and prepare a circular trench 1.0 m wide and 15 cm deep around the tree. Incorporate 30 kg of biochar with 20% moisture content into the soil in the trench.

In the multiple application method, maintain the same 1.0 m distance from the palm base and prepare a trench 1.0 m wide and 15 cm deep in one-fifth of the manure circle. For the first application, incorporate 6 kg of biochar with 20% moisture content into the soil in the trench. Repeat this process either annually for five consecutive years or five times in the same year, progressively covering the entire manure circle without overlapping previous applications. This approach ensures gradual soil enrichment and enhanced nutrient distribution.



Biochar application for adult Coconut palms

After the application of biochar, all the other recommended agronomic practices should be followed (e.g, fertilizer application and mulching).

Irrigation for Coconut Cultivation

Factors to be Considered When Adopting Irrigation Practices in a Coconut Plantation

1. Why is irrigation necessary?
2. How much water should be supplied?
3. How should water be provided?

Why Irrigation is Necessary?

Water is essential for both vegetative growth and nut production in coconut palms. During dry seasons, insufficient soil moisture reduces the nut yield, decrease nut size, and lower the copra quality. Therefore, supplemental irrigation is essential to sustain optimal palm growth and productivity.

Water Requirement of Coconut Palm

Mature coconut palms require 40 - 60 L of water per day. This varies greatly according to prevailing environmental conditions.

E.g.: this value may increase up to 100 L of water per day due to increased evaporation in mature coconut palm under high heat stress conditions.

The water requirement of the young coconut palms varies with age.

Water requirement at different ages of the coconut palm

Age	Water requirement (L/day)
Less than 1 Year	12
1 Year	18
2 Years	28
3 Years	32
4 Years	36
5 Years	40-60

Water requirements vary according to climate zones.

- Intermediate Zone : Rainfall is less than the water requirement
- Dry Zone : Rainfall is less than the water requirement
- Wet Zone : Rainfall in the wet zone is sufficient for coconut cultivation

Note: Even in the wet zone, evenly distributed rainfall throughout the year is essential. If rainfall is uneven or prolonged dry periods occur, irrigation is required to sustain coconut yields.

Key factors in selecting an irrigation system for coconut cultivation

1. Water source and quality
2. Soil factors
 - Soil texture
 - Soil aeration
 - Field capacity
 - Water holding capacity of soil
 - Drainage
3. Active root zone of coconut palms
4. Economic factors
5. Intercropping
6. Labor requirements
7. Nature of the Land



Agricultural ponds

Methods of Irrigation: Traditional and Improved Methods

Traditional Methods

1. Using small agricultural ponds

- In this method, water is taken manually from small ponds in the field using buckets or similar containers for irrigation.
- This is a very simple method.

2. Pitcher irrigation

- Pitcher irrigation slowly releases water to coconut palms through unglazed clay pots buried on either side of the palm.
- The pots contain small pores that allow water to seep into the surrounding soil, while the side facing away from the palm is painted to minimize water loss. A 1-meter-long coir rope can be attached to the bottom of the pot and buried toward the root zone to enhance water use efficiency.
- This method is especially suitable for young palms in sandy soil and requires manual labour to fill the pots.
- Weed growth is minimal due to the localized watering.



Pitcher irrigation for young palm



Modified pitcher for pitcher irrigation

3. Basin irrigation

- Basin irrigation supplies water to coconut palms using shallow, round basins built around each tree.
- These basins are connected by ditches to direct water flow, except in sandy soils.
- Water is released from a source at one end into the ditch system, allowing it to flow toward the basins.
- Suitable for irrigating mature coconut palms.
- More effective on flat or gently sloping land.
- Requires a large water source due to high water wastage.
- Movement of machinery is difficult in fields with basin irrigation.



Basin irrigation in coconut plantations

Improved Methods

1. Sprinkler irrigation

- Sprinkler Irrigation uses pumps, pipes, and sprinklers to distribute water evenly across the land.
- This method is suitable only for irrigating seedling nurseries and coconut plantations with intercrops.
- Requires relatively low labour input.
- Not suitable for areas with heavy winds.
- Requires a significant capital investment for installation and maintenance.



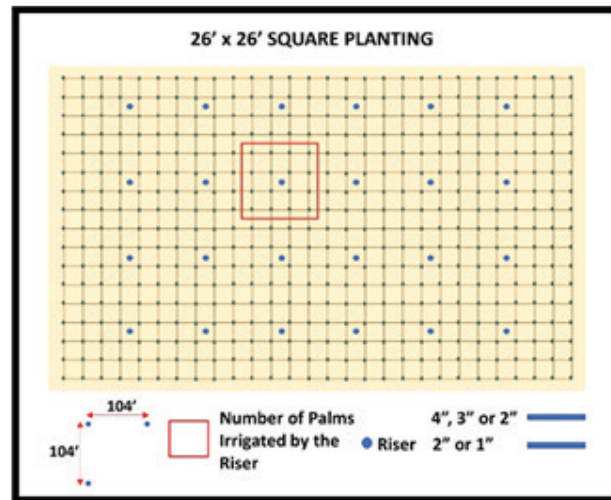
Sprinkler irrigation method

2. Hose irrigation

- Hose Irrigation uses underground pipes and risers with control valves to supply water to palms through long rubber hoses.
- Typically, one riser is used to irrigate 16 palms in an 8 m × 8 m square planting system, using a 30 m long rubber hose with a 32 mm to 63 mm diameter.
- This method requires labor to manually handle and operate the hoses.
- Simple to use and does not require special technical knowledge.
- Easy to maintain, with lower maintenance costs compared to other methods.



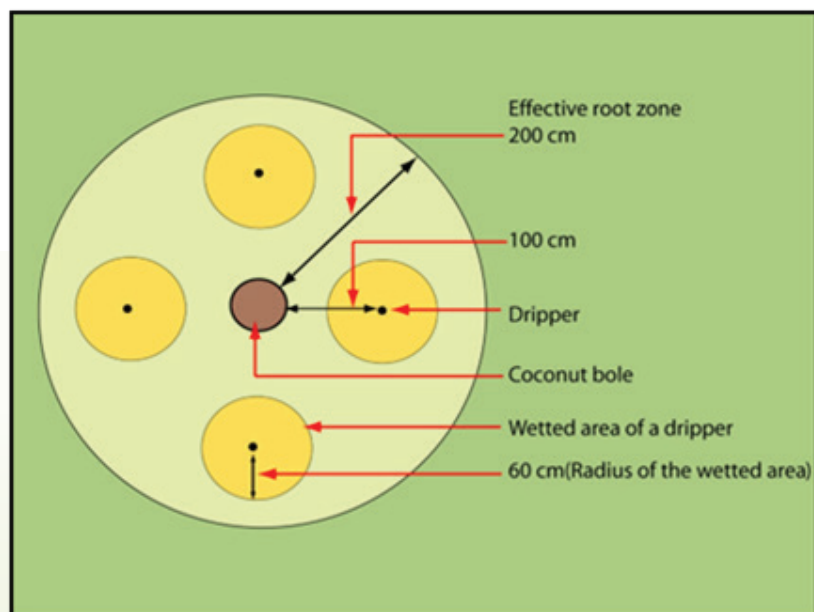
Hose irrigation method



Hose irrigation system layout in a square planting system

3. Micro irrigation

- Micro-irrigation supplies water in small amounts through emitters placed at several points within the manure circle.
- This method ensures high water use efficiency due to minimal water wastage.
- Requires significant capital investment for installation and maintenance.
- Requires relatively low labour input
- Good quality water is essential to prevent clogging and ensure long-term system performance.
- There are two methods of micro-irrigation: surface micro-irrigation and sub-surface micro-irrigation.
- Wetting patterns of soil differs with the texture of the soil and water flow rate from the emitters. Accordingly, the best way to set up the drip emitters, the speed of water flow for different soils, time to release the water, and the intervals to supply water have been introduced.



Establishment of emitters around the coconut palm

The irrigation schedule suggested for adult palms

Soil type	Discharge rate of water per emitter (L/hr)	Duration of Irrigation (hr)	Irrigation interval (days)
Sandy clay loam soils with gravel	30	2	4-6
	15	4	4-6
	8	8	4-6
Deep sandy loam soils	15	4	4-6
	8	8	4-6
	4	8	2-3
Moderately deep to deep sandy soils	4	8	2-3
	2	8	Daily

The irrigation schedule suggested for young palms

Age of the palm	Discharge rate of water per emitter (L/hr)	Duration of Irrigation (hr)	Irrigation interval (days)
<1 year	30	0.5	5
	15	1	5
	8	2	5
1 year	30	1	6
	15	2	6
	8	4	6
2 years	30	1.5	6-7
	15	3	6-7
	8	6	6-7
3 years	30	1.75	6-7
	15	3.5	6-7
	8	7	6-7
4 years	30	1.9	6
	15	3.7	6
	8	7.5	6
5 years	30	2	6
	15	4	6
	8	8	6

Surface micro-irrigation

- Surface micro-irrigation provides water through small pipes or emitters placed on the soil surface around the coconut palm.



Surface micro-irrigation

Subsurface micro-irrigation

- Subsurface micro-irrigation supplies water through pipes or emitters buried below the soil surface, directly to the roots of the coconut palm.
- Subsurface micro irrigation can prevent damage to the pipe system caused by machinery.



Subsurface micro-irrigation

4. Girdle Sprinkler Irrigation

- Girdle sprinkler irrigation wraps a small pipe with holes around the palm to slowly water it.
- This girdle pipe is connected to an underground pipe system.
- This method addresses several practical limitations of conventional drip irrigation systems.
- Capital cost is lower compared to drip irrigation.
- The water-use efficiency of this system is about 75 – 85%.



Girdle irrigation method



Pest and Disease Management in Coconut Plantation

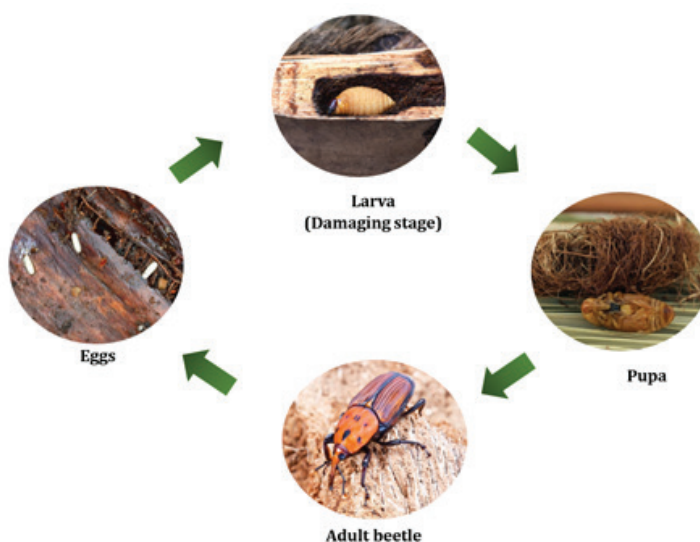
Major Pests in Coconut Plantations

- Pests and diseases in coconut plantations can lead to severe damage, significantly reducing crop yield and, in some cases, causing death of palms.
- If not addressed promptly, these issues can escalate, affecting both productivity and long-term plantation health.
- Therefore, timely and effective management interventions are essential to safeguard the coconut plantations.
- Integrated Pest and Disease Management is a comprehensive approach to maintain plant health by minimizing the impact of pests and diseases through the strategic combination of multiple control methods.
- This involves the integration of cultural, biological, mechanical, and chemical control strategies to achieve sustainable pest and disease management.
- The effectiveness of this approach depends on the synergistic application of two or more management practices.
- This holistic approach enhances the resilience of coconut palms and reduces the risk of resistance development in pests and disease-causing organisms, and minimizes adverse environmental impacts.

1. Red Palm Weevil (*Rhynchophorus ferrugineus* Olivier)

- This pest severely damages the palm from trunk emergence up to 15 years.
- Damage is caused by the larvae.

Lifecycle of the Pest



The lifecycle of red palm weevil consists of 4 stages; namely, egg, larva, pupa and adult. The adult female deposits eggs on the wounds on the trunk, petioles and exposed roots of coconut palms. The eggs are elongated (about 2.5 mm) and creamy white in colour. Larvae that are hatched from the eggs are whitish or yellowish with a large hard brown head and no legs. The hatched larvae then bore into the internal tissues of the palm trunk and progress through several instars. Their size ranges from 2 mm upon hatching up to 5 cm at full growth (in length). The last larval instar becomes a pupa, which is protected in a cocoon made of palm fibers. The cocoons are formed under the bark of the stem or petiole. The pupa inside the cocoon is immobile; therefore, it does not crawl anymore. Adults who emerge from the pupae are reddish brown with black spots on the thorax. They bore holes in the stem and leave the palm.

Characteristics of the Damage

- Stems with 1-2 cm diameter holes with fibres coming out of the holes or oozing brown liquid.
- Yellowing and drooping of leaves.
- The presence of fibrous cocoons at the bases of petioles and beneath the bark.
- A crunching noise is heard when larvae feed inside the trunk, if the ear is placed close to the damaged trunk Presence of small holes (0.5 - 1 cm in diameter) at the places where the larvae have entered, when the petioles are removed.
- If the pest is not effectively managed, under infestation, the crown of palm topples down.



Crown toppled palm due to red weevil damage

Control Methods

- An integrated pest management is essential to control this pest.

1. Agronomic Methods

- Care should be taken to avoid any injury to the palms.
- Susceptible palms should be thoroughly inspected every 2–3 weeks.
- Apply coal tar or burnt engine oil (engine oil removed from vehicles) to the wounds, caused by damages to the tree trunk (mechanical damage, damage caused by cattle, natural damage due to overgrowth of trunk in areas with high water table), bark (black beetle damage), crown (splitting during drought) and roots (mechanical damage to exposed roots or animal damage).
- During drought, when the branches of the trees are drooping due to the pulling of cattle, first cut the branches with a sharp knife and apply coal tar or burnt engine oil on the wounds where the fronds are attached to the trunk.
- Severely damaged palms, which are beyond recovery, should be cut, split and burnt. Also, uproot the palm completely, split and burn.

The gnawing of red weevil larvae inside the trunk can be heard clearly using a red weevil detector, facilitating early identification of damage.

2. Chemical Methods

- Inject systemic insecticides such as Monocrotophos 600 g/l SL or an insecticide containing a mixture of Chlorantraniliprole 20% + Thiamethoxam 20% WG into palms when damage is initially identified.
- i. **Application of granular insecticide (mixture of Chlorantraniliprole 20% + Thiamethoxam 20% WG)**
 - Dissolve 4 g of the insecticide powder in 250 ml of water to make a solution and inject into the trunk.
 - o 60 ml of the above solution should be injected for each palm with a trunk height of 1.5 m or less
 - o 120 ml of the above solution should be injected for each palm with a height of more than 1.5 m (in two levels)

Method of Applying Insecticide

- This insecticide is in powder form.
- First, fill a bottle with the insecticide (4 g packet), add 250 ml of water, and shake well to dissolve the powder.
- Based on the height of the palm, make 10 mm deep and 13 mm diameter holes using a manual drill or electric drill at a downward 45° angle in the stem.
 - o If the palm is 1.5 m tall or less, make the first hole 30 to 60 cm above ground level and the second hole about 1.2 m above ground level. If the palm is taller than 1.5 m, drill two more holes on either side of the 1.5 m mark.
- Inject the necessary amount of insecticide into the holes.
- When applying insecticides, divide the volume required for a palm by the number of holes created and then apply in equal amounts.
- One week after applying the insecticide, seal the holes made by the damage caused by the Red weevil and the insecticide application sites using a mixture of sand and cement.
- If symptoms such as oozing persist even after two weeks from applying insecticides, make new holes and proceed with the insecticide application as above.
- If external symptoms can approximate the location (newly formed ooze holes) of the larvae in the stem, injecting the insecticide with one hole above and below that area is more effective.
- Since insecticides are toxic, take adequate precautions when using them.

ii. Injecting Monocrotophos 600 g/l SL Systemic Insecticide

- Monocrotophos dosage:
 - o For a palm in which the trunk is not formed – 10 mL in 500 mL of water
 - o For a palm with a thin trunk (circumference at the base of the trunk is less than 100 cm) – 30 ml
 - o For large trunk palms (circumference more than 100 cm) – 40 mL
- For mature palms, inject insecticide into the trunk or allow the insecticide to be absorbed from the root into the palm.
- If the trunk of the young palm has not yet formed, thoroughly wet the bud region by drenching a mixture of 10 ml of Monocrotophos 600 g/l SL insecticide in 500 ml of water into the crown.
- Before applying the insecticide, 2 mature coconut bunches should be harvested from the damaged palm, and the coconuts from such palms should not be consumed for 2 months after applying the insecticide.

Trunk Injection

- Using an electric drill or a manual drill, prepare holes approximately 13 mm in diameter and 50 cm – 100 cm above the ground level of the palm, at an angle of 45° from opposite directions of the trunk. Drill two holes about 10 cm deep each and remove the fibre parts within those holes.
- Apply 50% of the recommended amount of insecticide per hole as a concentrated solution (without adding water) using a funnel or syringe. If the entire amount cannot be applied at once, the rest can be applied after about an hour.
- One week after treatment, cover the hole with a mixture of sand and cement and apply coal tar on it.
- Mark the treated palm for easy identification.



Trunk injection of Monocrotophos 600 g/l SL

Root Feeding of Insecticide

- Carefully dig the soil at the base of the trunk and select a light brown, healthy root about the thickness of a pencil.
- Mix the required dose of Monocrotophos 600 g/l SL in an equal volume of water and pour half of the mixture into a polythene bag (about 6 x 15 cm) made of thick polythene or a glass tube or a small bottle.
- Cut the tip of the selected root into a slanted shape with a sharp knife and insert the cut end into the bottom of the bag containing the Monocrotophos 600 g/l SL mixture and tie the open side of the bag tightly to the root.
- Tamp the root into place or cover it with another cover.
- Repeat the same procedure to another healthy root with the rest of the insecticide solution.
- Mark the insecticide-treated palm for easy identification.
- Harvest two mature coconut bunches from the damaged palm before treating with insecticide and coconuts from such palms should not be consumed for 2 months after applying the insecticide.



Root feeding of insecticides

Use of Pheromone Traps

- The pheromone trap can be used when red weevil adults are found in the field.
- The pheromone trap should be used very cautiously. Its use is recommended only for coconut plantations of more than 5 acres.
- Parts of the pheromone trap
 - o 5 – 6 l plastic bucket
 - o Pheromone tube (pheromone vial) or gel-based pheromone
 - o A small container of bait (only when using the pheromone tube)



Lid

Pheromone lure (vial with bait/gel- based pheromone)

Bucket (1/4 filled with soap water)

Parts of the Red weevil pheromone trap

- The Coconut Research Institute of Sri Lanka has introduced two types of pheromone lures.
 - o Gel-based pheromone
 - o Pheromone Vial



Pheromone Vial



Gel-based pheromone

1. Gel-based Pheromone

- This is an improved pheromone lure that does not require bait.
- When installing the trap, remove the gel pheromone sachet from the outer cover and tie it to the support attached to the upper rim of the bucket.
- Pour a soap solution to about 7.5 cm into the bottom of the plastic bucket. Make sure to maintain the water level by refilling it when the soap solution dries out.

2. Pheromone Vial

- When using pheromone vials, pour a soap solution about 7.5 cm deep into the bottom of the plastic bucket.
- Refill the soap solution as needed.
- Fix the baited bowl near the rim of the container.
- Use 100 ml of toddy or yeast-sugar solution as the bait. Yeast sugar solution: 100 g sugar + 2 g yeast + 1 L water
- Carefully break off the top (thin side) of the pheromone vial and tie it to the support attached to the top of the bucket.
- For successful control of the red weevil, five pheromone traps should be installed per hectare (one trap per 0.5 acre).

- The traps should be installed at different places in the plantation.
 - o Fix the trap to a stick at the corner of the plantation to prevent it from being blown by the wind.
 - o Clean the trap every two weeks and refill with soap solution and bait. Leave the traps in the field until the number of beetles caught drops to fewer than three per month.
 - o Replace with a new pheromone vial or gel-based pheromone when the previous one runs out of chemicals.
 - o More effective results can be achieved if the traps are installed simultaneously in a large area (village level).
 - o The pheromone trap should be installed in a mature plantation adjacent to the infested land or at the edge of the field.



Pheromone trap installed in coconut plantation

2. Coconut Mite (*Aceria guerreronis* Keifer)

Identifying the Pest

- Invisible to the naked eye (microscopic).
- It lives under the perianth of the nut and sucks the sap from soft tissues.
- Nuts of all ages are susceptible to mite damage.
- A large number of mites inhabit nuts between 3 and 7 months old.
- Spreads rapidly via wind, animals, insects, and human activity.



Microscopic view of the pest

Symptoms of Mite Damage

- A yellowish-white triangular spot first appears below the perianth.
- This spot gradually enlarges and turns brown with the growth of the nuts due to continuous sucking of sap by the pest.
- Cracking and drying of the nut surface cause the nut to become small, distorted, and ruptured, reducing quality. Two other mite species (*Dolichotetranychus* and *Colomerus* mites) have been identified, and it is important to distinguish their damage, which is not economically significant.
 1. A wide brown ring appears around the nut when damaged by *Dolichotetranychus* mite.
 2. An elongated, brown, square-shaped patch appears on the nut when it is damaged by *Colomerus* mites.



Mature coconuts damaged by the pest



Different stages of symptoms of the coconut mite

Control Methods

It is difficult to eliminate mites from palms.

- However, the damage can be reduced following the control measures mentioned below.

i. Chemical Control

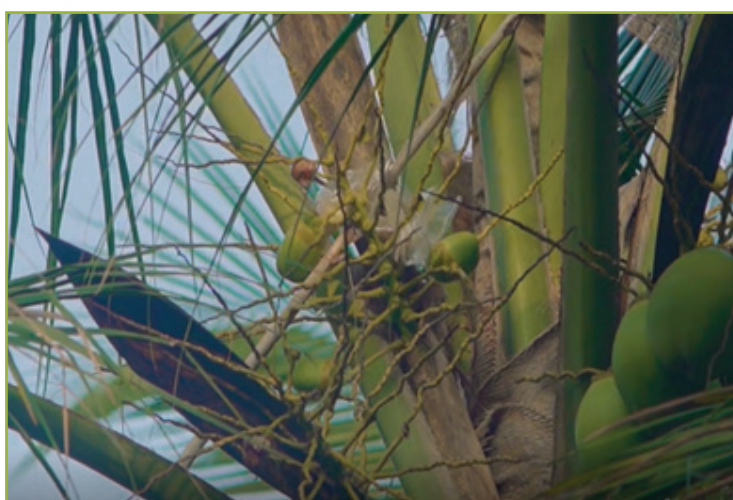
- Spraying a mixture of palm or vegetable oil and wettable sulfur.
 - Preparation of 1 L of mixture
 - o 200 ml of palm/vegetable oil (1 cup)
 - o 800 ml of water (4 cups)
 - o 12 g of soap powder (2 full tablespoons)
 - o 5 g of sulfur powder (water-soluble) (1 full tablespoon)
- First, dissolve soap powder and sulfur in water. Then, add palm/vegetable oil and mix well.
- By attaching a long plastic tube to a knapsack sprayer, this mixture can be sprayed up to 12 m tall with the tank on the ground. Alternatively, a power sprayer can also be used. This method can destroy more than 90% of coconut mites.
- It is important to apply the mixture to the crown area, ensuring complete coverage of all mite-infested coconut bunches.
- About 1 L of the mixture is sufficient for one palm.
- It is recommended to spray this mixture twice a year, particularly during February and from July to August.
- After spraying, the damaged area will separate from the perianth, and the damage will stop. Infestation gets minimized in the newly developing button nuts.
- This mixture will destroy only *Aceria* mites, and not any other predatory mites.



Spraying for management of mite

ii. Biological control

- A predatory mite (*Neoseiulus baraki*) has been recommended by the Coconut Research Institute for controlling the coconut mite.
- This predatory mite species is found locally.
- Although complete coconut mite control cannot be achieved, crop loss can be significantly reduced.
- The release of 5,000 predatory mites in one-fourth of the mite-infested plantations three times a year is recommended.
- Predatory mite packets can be purchased from the Coconut Cultivation Board.
- The predatory mites can be purchased in polypropylene sachets. They contain predatory mites with a food source and moist tissue paper to keep the relative humidity inside the sachet high.
- In the field, cut open the end opposite the tissue paper and carefully place the sachet between the immature bunches on the palm.
- After about 2 days, all predatory mites will have left the sachet and settled under the perianth of the nuts.



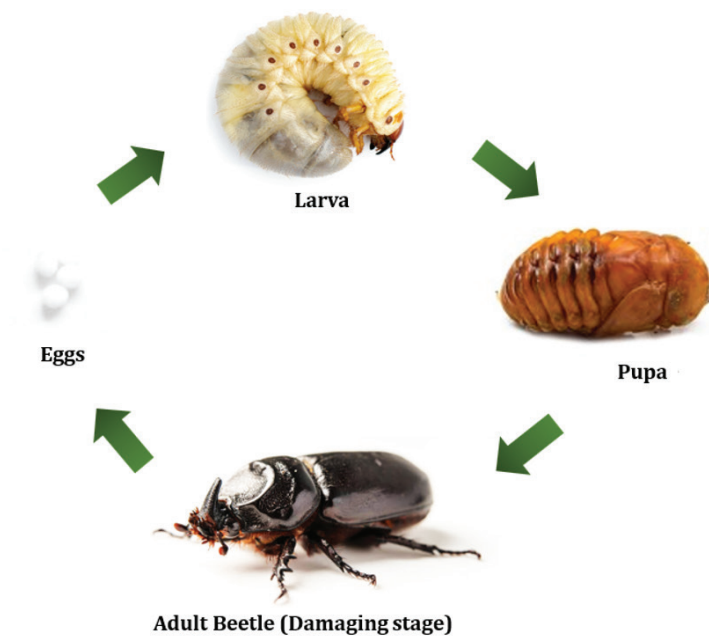
Placing a sachet with predatory mites on an immature coconut bunch using a pole

3. Black beetle (*Oryctes rhinoceros* Linnaeus)

- Black beetle damage is found in almost all areas of the country.
- Palms of all stages are susceptible to damage by the black beetle.
- The damage stunts the growth of the palm, and severe damage to young palms could be lethal.
- Damage is caused by adult beetles.

Lifecycle of the Pest

The lifecycle of the black beetle consists of four stages: egg, larva, pupa and adult. The adult female lays eggs on decaying organic matter. These eggs are globular in shape and white in color. The larvae hatched from the eggs are creamy white in color and have a characteristic C- shape. They feed on decaying organic matter. After several instars, the larvae become pupae and then adults. Adults are black in color and have a characteristic horn-like structure at the anterior end (head of the beetle).



Black Beetle Breeding Sites

Black beetles breed on decaying coconut trunks, basal parts of fallen coconut palms, cow dung pits, coir dust, paddy straw, off-cuts from coconut rafter processing areas, and organic manures."

Identification of the Damage

- This beetle bores into and enters the soft areas at the base of the bud and feeds on soft tissues.
- New damage can be easily detected by chewed fibers discarded at the base of the palm or around holes where the beetles have entered.
- Due to their feeding, unopened coconut leaves and petioles are damaged.
- These leaves appear cut in a "V" shaped pattern when unfolded.
- When the petiole is damaged, the flag leaf breaks down.
- Young palms can develop deformed fronds, and if the growing bud is damaged, the plant will die.



Black beetle damage in a young palm



Symptoms of Black beetle damage

Control Methods

- An integrated pest management practice should be adopted for black beetle control.

i. Agronomic Methods

- Burnt engine oil should be applied to coconut logs before using them for poles or other purposes.
- Remove piles of wood shavings and wood chips in the coconut plantation. Destroy them if not needed or thin them out to less than 5 cm if needed for a specific purpose.
- Do not leave manure piled in the field for more than 6 months.
- When using organic matter as fertilizer, mix it thoroughly with the soil.
- When using coir dust as mulch around palms, spread it into a very thin layer.
- In cases where coconut fronds are used as mulch, check them for black beetle larvae as the fronds decompose. If larvae are present, they should be destroyed.
- A soil layer of more than 15 cm must be maintained on the surface of coconut husks or coir pits, which are used for moisture conservation in coconut cultivation. During planting, tie the polybag without removing it completely to cover the base of the plant. It is advisable to make some holes in the polybag for ventilation.

ii. Mechanical Method

Removal with a pointed metal hook

- A metal hook prepared like an arrow can be inserted into the hole where the black beetle entered, and the beetle can be hooked out.
- Then apply coal tar or burnt engine oil around this hole.



Removing a live adult black beetle feeding inside the young palm using a metal hook

Use of fish nets

- Wrap a 30 cm to 90 cm piece of fish net with small holes to cover the area where black beetles are likely to enter the young plants. Adjust the net piece back up once every 2 months as the plant grows. This acts as a mechanical barrier protecting them from beetle attacks. Tuck a piece of mesh between the petiole as well. It can be used successfully until the plant becomes 3-4 years of age.



Wrapping the bud region using a piece of fishing net (netting) to block the black beetle entry into the bud

iii. Chemical Methods

- It is important to apply the following measures once a month to control the black beetle damage.
 - o Apply coal tar or burnt engine oil on the coconut petiole around the bud.
 - o Put 6 naphthalene balls per young palm, at the bases of the petioles around the bud.

Use of Pheromone Traps

- One pheromone trap is sufficient for 5 acres of plantation.
- This pheromone attracts adult beetles (Aggregation Pheromone).
- Best suited for areas with high pest damage.

Preparation of the Pheromone Trap:

- The pheromone trap is prepared using a 1.8 to 2.0 m, 10 to 12.5 cm PVC pipe (Down pipe).
- Two square-shaped windows should be cut opposite to each other as shown in the figure (for entry of beetles).
- Drill a hole 1.5 cm above one window in this tube (to hang the pheromone).
- Drill 4 holes on both sides of the bottom end of the tube (to drain rainwater).
- Attach a lid (end cap) to the bottom edge of the holes that have been drilled to cover the holes well.
- Fill this tube with coir dust (2 - 3 kg) (as a medium for trapped beetles).

Installing the pheromone trap in the Field:

- The above trap can be installed in the field with the help of pegs or gliricidia sticks or it can be tied to the trunk of an adult palm.
- The pheromone trap should be installed in a mature plantation adjacent to the infested land or at the edge of the field.



Pheromone Trap in the Field

Maintenance of the Pheromone Trap:

- After 7 - 10 days of installing pheromone traps, the beetles get trapped inside the pipe. Beetles trapped should be removed at about 10-day intervals by removing the lid and discarding the trapped beetles.
- Place the removed coir dust back inside the pipe.
- Destroy removed beetles without releasing them back into the field.
- Generally, the pheromone remains in the field for about 2-3 months (but this period may vary depending on the climatic conditions). If the pest damage is still present, the pheromone trap should be installed again in the field.

4. Coconut Leaf-eating Caterpillar (*Opisina arenosella* Walker)

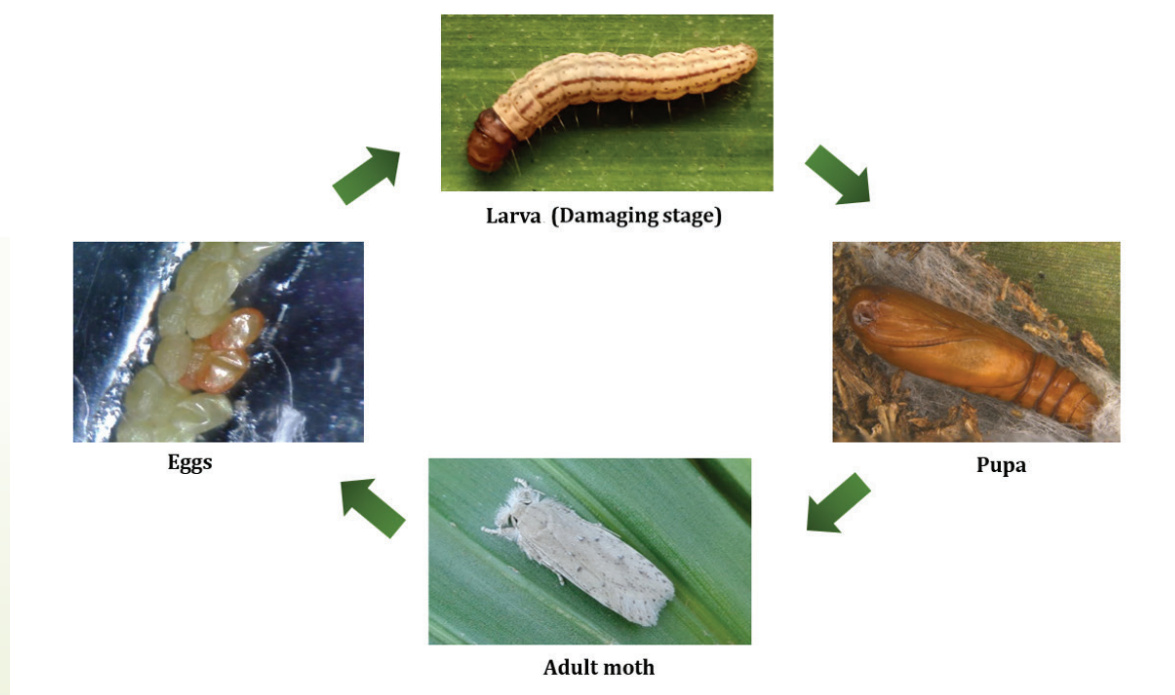
- Epidemic conditions can be found mostly during May to October.
- The damage is caused by larvae (caterpillars).



Coconut caterpillar (larva) and its gallery

Lifecycle of the Pest

The Lifecycle of the coconut leaf-eating caterpillar consists of 4 stages: eggs, larvae, pupae and adults. The eggs are laid on the undersides of coconut fronds in the lower whorl of the palm. The caterpillars that hatch from the eggs cause the damage. The larvae are pinkish when hatched and gradually turn to a greenish brown color with development. The head is brownish black in color and has dark brown longitudinal stripes along the dorsal surface. They form galleries and live inside. The larva then develop into pupa that are covered with a cocoon. The adult that emerges from the pupa is a brownish moth.



Characteristics of the Damage

- The caterpillars feed on the green tissue on the underside of the coconut leaflets.
- These areas first turn into light green and then to brown.
- The infestation can be identified by dry patches on the leaves.
- The infestation begins at the lower fronds of the coconut tree. On an infested frond, galleries made by the caterpillar from small pieces of leaves and their frass can be seen under the coconut leaves.
- When these galleries are opened, live caterpillars or cocoons can be seen.
- The palms will have a burnt appearance in severe infestations.

Control Methods

- If caterpillar numbers are low, natural enemies (parasitoids) in the field will naturally control the infestation.



Coconut leaf eating caterpillar damage

i. Mechanical Method

- When the infestation starts in a few palms (about 30 trees), cut and burn only the infested fronds, without exceeding five fronds per palm or the tips of the fronds. Alternatively, the tips of the fronds can also be cut and burned if the caterpillars present on tips.

ii. Biological Method

- Release parasitoids if the parasitoid population in the field is insufficient. For this, the growers are advised to contact the Coconut Development Officer in their areas who will come and inspect the infested lands for population counts of the pest and the parasitoids. These counts will be forwarded to the Coconut Research Institute, and based on the level of infestation, life stage of the pest and the presence/absence of the parasitoids, the Coconut Research Institute will start releasing the parasitoid. The growers are advised to contact the Crop Protection Division of the Coconut Research Institute, once the population counts are forwarded to the Coconut Research Institute. Parasitoids will either be sent to the estate via normal post, or the growers can come and collect them from the Crop Protection Division with a prior appointment. If the population is so high that the parasitoids alone cannot control them, pesticides will be applied first, followed by the release of parasitoids. However, this recommendation will be given by the Coconut Research Institute only after studying the populations of pests and parasitoids on the infested palms.

iii. Chemical Methods

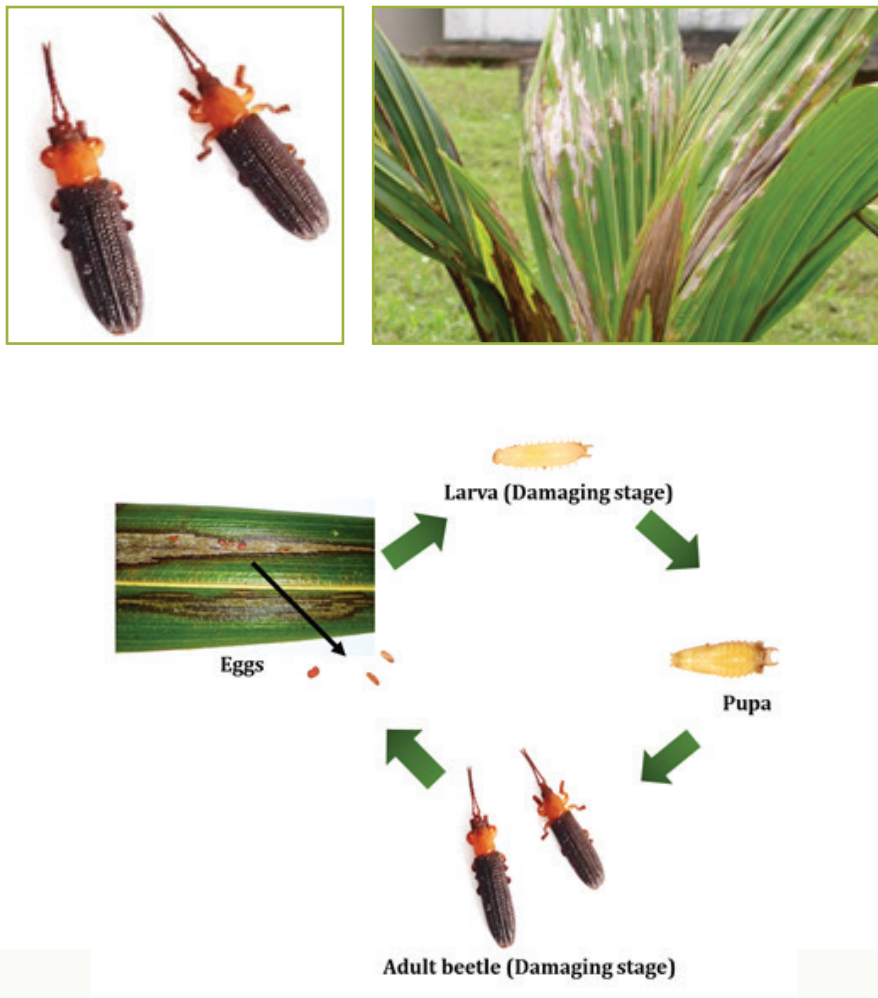
- For young plants, dilute 4 ml of an insecticide containing Carbosulfan 200 g/l SC in 1 L of water and spray the affected fronds to thoroughly wet the fronds.
- In case of an epidemic situation, consider the injection of Monocrotophos 600 g/l SL (8 ml per palm), which is recommended by the Coconut Research Institute.

5. Plesispa Beetle (*Plesispa reichei* Chapuis) (Coconut Hispid)

- Damages coconut seedlings in nurseries and newly planted ones in the field.
- The damage is done by both larvae and adults.

Lifecycle of the Pest

The lifecycle of *Plesispa reichei* consists of 4 stages: namely, eggs, larvae, pupae and adults. The adult female lays eggs in the bud region, on unfolded leaves. The larvae hatch from the eggs, develop through several instars that damage the leaves by feeding. The last instar larvae become pupae and then develop into adults. Adults have a brown thorax and black elytra.



Characteristics of the Damage

- Adults and larvae that live in the bud region scrape away the leaf tissue.
- Affected areas turn brown and dry.
- Due to the damage, the plant growth is stunted, and in severe infestations, the seedling may die.
- Damaged leaves may tear off during strong winds.

Control Methods

- Dissolve 3-4 ml of insecticide containing carbosulfan 200 g/l SC in 1 L of water and spray the bud region thoroughly.
- If infestation persists, spray insecticide again after 2 weeks as recommended.

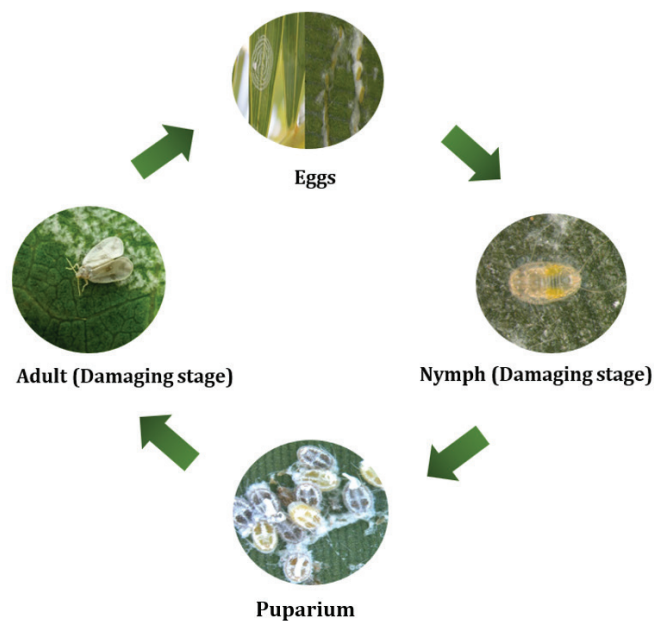
6. Whitefly

- There are five species of whiteflies associated with coconut in Sri Lanka. The reported species of whiteflies in coconut in Sri Lanka are *Aleurodicus rugioperculatus*, *Paraleyrodes minei*, *Aleurotrachelus atratus*, *Aleurodicus disperses* and *Paraleyrodes bondari*. It has been observed that various varieties of coconut palms and king coconut palms of all age groups are susceptible to damage by whiteflies. Apart from coconut and king coconut, many other crops such as banana, chillies, papaya, guava, black pepper, cucumber, curry leaves, mango, Indian-almond, and jackfruit etc, are also infested by this pest.



Whiteflies and their white, cottony nests on the underside of coconut leaflets

Lifecycle of Whitefly



Though five different species of whiteflies co-exist in coconut palms, they all have the same stages in their lifecycle. The stages are eggs, nymphs and pupae (immature stages) and adults. The egg laying pattern, egg colouration, shape and the size of eggs vary depending on the species. The nymphs that emerge from eggs are transparent and capable of crawling. Pupae are sessile and have considerable differences among species. Adults of all whitefly species associated with coconut resemble tiny moths. However, the size, shape, and pattern of pigmentation on wings differ among species.

Whitefly-damaged coconut or King Coconut palms exhibit a whitish, fluffy appearance under the leaflets. Upon closer inspection, various life stages (eggs, nymphs and adults) can be observed.

Their metabolic processes produce sugary substances known as honeydew, which are excreted from the tip of their body. When the drops of honeydew are deposited on the leaves below, black colour mold appears (Sooty mold) on whitefly-infested coconut leaves. If the honeydew is deposited on other plants growing under the coconut, the sooty mold can grow on those plants as well. In cases of high whitefly population, they can be found on coconut petioles as well as on nuts.



Growth of black sooty mold that reduces/ blocks photosynthesis

Control Methods

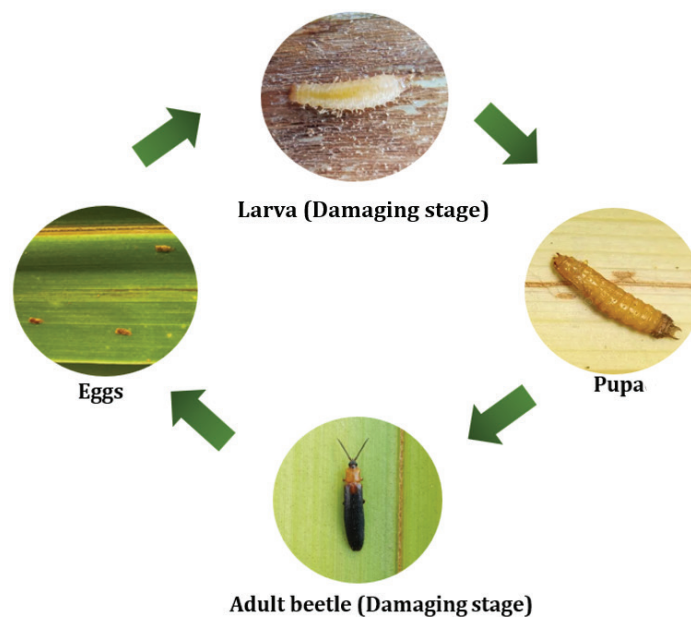
- The Coconut Research Institute has recommended the following insecticides as interim recommendations for the control of whitefly.
- The following synthetic insecticides are recommended for new infestations/outbreaks in areas and for nurseries.
 - o Dissolve 3 g of Thiamethoxam 25 WG in 10 L of water and spray.
 - o Dissolve 20 ml of Carbosulfan 200 g/l SC in 10 L of water and spray
 - o Dissolve 2.5 g of Chlorantraniliprole 20% + Thiamethoxam 20% WG in 10 L of water and spray
- In addition, the following mixture is recommended for controlling whitefly in any coconut-growing area of the country.
 - o Mix 10 ml of neem oil and 5 g of soap powder in 1 L of water and spray thoroughly under the leaves infested with whiteflies. This mixture should be sprayed once every 2 weeks for at least 2 months to get effective results. Growers may contact the Coconut Development Officer in their area to get assistance on power sprayers to spray the chemicals.
- The whitefly damage is naturally controlled to some extent by parasitoids and predators available in the field, and damage is reduced after heavy rains. Also, the damage can be reduced by applying smoke in coconut plantations by burning trash in the plantations without damaging the palms by fire.
- Apart from this, applying grease to a yellow polythene sheet and wrapping it around the trunk of coconut palms or King Coconut palms can suppress the adult population of whiteflies to some extent.

7. Brontispa Beetle (*Brontispa longissima* Gestro)

- It is a newly identified invasive pest in coconut cultivations in Sri Lanka.
- Currently, this infestation is reported only from the Galle, Matara and Hambanthota districts of the Southern Province.
- In case of an epidemic situation, the coconut plantation is severely damaged.



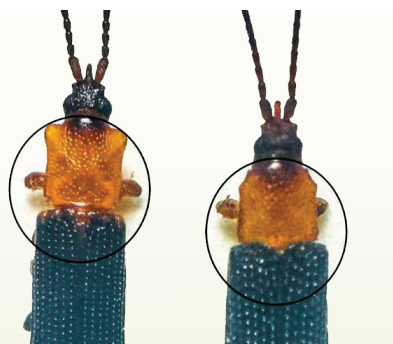
Lifecycle of Brontispa Beetle



The lifecycle of *Brontispa longissima* consists of 4 stages: eggs, larvae, pupae and adults. The adult female lays reddish-brown eggs in the bud region, on unfolded leaves of adult palms. The yellowish larvae hatch from eggs, develop through several instars that damage the leaves through their feeding activity. The last instar larvae become bright yellow pupae and then develop into adults. Adults have an orange thorax and black elytra (forewings). They also feed and live in the bud region.

Identification of Brontispa Beetle

- This beetle can be identified by its black head and elytra (upper wings), and characteristic orange colored, square-shaped thorax.
- They are morphologically very similar to the *Plesispa reichei* found in coconut plantations. Compared to the *Plesispa reichei*, *Brontispa longissima* beetle causes more severe damage to coconut plantations.
- This pest attacks any stage of coconut cultivation and particularly causes severe damage to mature coconut palms.
- The adult and larval stages of this insect feed on the immature tissue of the unopened leaves in the bud region.
- After the leaves unfold, the leaves appear withered.
- As a result, the growth of the palm and the yield are reduced.



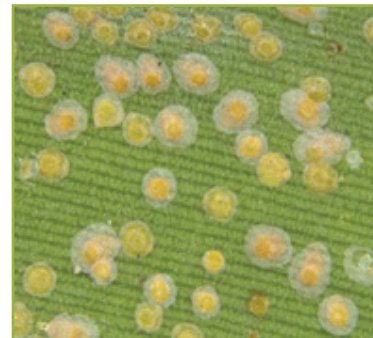
Control Methods

- Inject 10 ml of Monocrotophos 600 g/l SL by drilling the stem (trunk injection).
- Can be controlled biologically by releasing parasitoids into the field.

Minor Pests in Coconut Plantations

1. Scale insects (*Aspidiotus destructor* Signoret)

- A small insect with yellow coloured center surrounded by a transparent cover.
- Found in all coconut-growing areas.
- Damage is severe from August to February.



Scale insects living and feeding on the underside of coconut leaflets

Characteristics of Damage

- The female and immature male stages of this insect attach under the coconut leaf and suck the sap.
- First, the affected area appears as yellow spots, and when the number of insects is high, the entire leaf turns yellow.
- Gradually, many leaves of coconut tree turn yellow and dry up.
- In cases of high insect populations, coconut flowers and fruits are also damaged.
- If the scale insects living under the coconut leaf are alive, they are felt wet when scraped with the fingernails, while if they are non-living, they are felt like a dry powder.

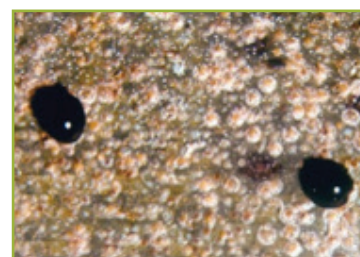
Control Methods

i. Mechanical Methods

- If damage to the coconut palms is slight, scrape off the scale insects by hand or remove the leaves.
- If only 4-5 lower fronds are damaged in a few palms, cut and burn the damaged fronds.

ii. Biological Methods

- Scale insects are destroyed/fed by two species of ladybird beetles that live naturally in the environment
 1. *Chilocorus nigritus* Fabricius
 2. *Pullus xerampelinus*



Chilocorus nigritus

iii. Chemical Methods

- If the pest is not controlled by biological control methods,
 - o Spray the young coconut plants with a solution of 2-4 ml of a Carbosulfan 200 g/l SC-containing insecticide in 1 L of water so that the plants are thoroughly wetted.

2. Termites

- The damage varies depending on the type of termites.

Characteristics of Damage

- A termite called "*Odontotermis*" causes severe damage to nursery seedlings and young plants planted in the field.
- Enters through the base of the seedlings or through the trunk and destroys the bud.
- In damaged seedlings, the bud turns yellow and dull, and it can be easily pulled out.
- Damage is high in clay soil.
- Another termite species damages the roots and base of mature palms.
- The outer bark near the base gradually peels off due to damage, and the palms decline rapidly.



Termites

Control Methods

- Demolition and removal of termitaries from the field.
- Keeping the land clean by burning decaying coconut fronds and unwanted nuts.
- Apply 3–5 ml of an insecticide containing Carbosulfan 200 g/l SC, mixed in 1 L of water, to affected nurseries and plants.
- Before planting, soak the seed nuts in the solution for 3 minutes.

3. Yellow Spotted Locust (*Aularches miliaris* Linnaeus)

- It is a common pest in forested areas of the wet zone.
- A fully-grown grasshopper is a large insect.
- The female lays eggs as pods in the soil.



Characteristics of Damage

- The pest feeds on coconut leaves, leaving only the ekel (midrib).
- They breed in large numbers and cause severe damage as groups.

Control Methods

- Destruction of eggs in soil by land preparation.
- Destroy the locusts hiding among the wild trees in the afternoon by following weed control methods
- Destroy locusts at night with a light trap.

4. Nettle Grub (*Parasa lepida* Cramer)

- The damage is done by the larval stage or caterpillar.

Characteristics of Damage

- The caterpillars feed on parts of the leaflets, leaving only the ekel.



Parasa lepida Cramer

Control Methods

- It is naturally controlled by larval parasitoids, pupal parasitoids and predatory insects.
- If the damage is severe, chemicals should be applied.
 - Spray young plants with 3 - 4 ml of an insecticide containing carbosulfan 200 g/l SC mixed in 1 L of water.
 - For mature palms, inject the trunk with 5 - 10 ml of Monocrotophos 600 g/l SL insecticide.

5. Elymnias Caterpillar (*Elymnias hypermnestra* Linnaeus)

- The caterpillar damages seedlings in the nurseries and young plants in the field.

Characteristics of Damage

- The female of the butterfly lays eggs on the leaves of coconut or other palm plants, and the caterpillars that hatch feed on coconut leaves.
- Leafy parts of coconut plants are fed completely as if it was cut with a pair of scissors.

Control Methods

- Destroy after hand-picking.
- Spray the leaves thoroughly to get wet with 3 - 4 ml of an insecticide containing Carbosulfan 200 g/l SC mixed in 1 L of water.



6. Leaf Curling Caterpillar (*Telicota palmarum*)

- Caterpillars damage seedlings and young plants just after planting in the field are resulting in stunted growth.

Characteristics of Damage

- The female lays eggs on the leaves of coconut or other palm plants.
- The caterpillars that hatch from the eggs take the leaf parts together, curl them up and feed on the leaf parts.

Control Methods

- Destroy after hand-picking.

7. Mammalian pests

- Mainly rats, greater bandicoot rats, porcupine, giant squirrel, wild boars and bats damage coconut plantations.

Rat

- Damage seedlings, young and adult palms



Rat damaged nuts

Characteristics of the Damage

- Due to digging up the basal part of the coconut palm and eating the young tissue parts, the bud withers/dies.
- The palm dies at any stage when the bud is damaged.
- Rats gnaw young nuts of bearing palms, feeding on the kernel and drinking the water. Varieties like Navasi and King Coconut are more susceptible to rat damage.

Control Methods

- By maintaining field sanitization (by weed control and clearing of palm crown).
- By placing baits in the plantation with rodenticides/rat traps placed on coconut trunks/ ground traps placed at ground level.
- Tie the trunk of palms with a 500-gauge or thicker, 30 wide polythene band.
- Tie the trunk with a surface smooth aluminum / galvanized sheet, which is 30 cm wide and long enough to wrap the palm around 1 m above the ground level.



Wrapping a galvanized sheet (width about 0.3 m) around the trunk

Greater Bandicoot Rat

- Feeds on the base of coconut seedlings in nurseries and coconut palms planted in the field.

Characteristics of Damage

- The bud withers and dries.

Control Methods

- Maintain field sanitization.
- Ground traps, ground baiting.
- Cover the base of young plants with chicken mesh or roofing tiles



Covering the base of the young plant with a chicken mesh to prevent bandicoot damage

Porcupine

- The base of the trunk in both mature and young palms is damaged, reducing their growth.

Control Methods

- Establish a fence along the estate boundary to safeguard the coconut land from forest animals. Wrap each plant's base with netting or install barrel halves to safeguard the plant base.



Placing a barrel (a mechanical barrier) covering the base of the young palm to prevent Porcupine damage

Bat

- Coconut trees near bat roosting sites are primarily affected.
- The damage is particularly severe during the dry periods.

Characteristics of Damage

- Feeds on button nuts, and damaged button nuts will fall.
- Sometimes the coconut leaves are torn and dried up when leaves on the top layer of the crown are damaged.

Control Methods

- Burning firecrackers in tree shelters during the day.
- This kind of practice prevents them from moving in that direction, at least for a short period.

Giant Squirrel

- Palms grown in areas close to forests are frequently damaged.

Characteristics of the Damage

- Destroys the button nuts in adult palms by feeding on them.

Control Methods

- Regularly inspect the plantation and expel them.

Monkey

- They drink coconut water and eat coconut kernel. In addition, button nuts and inflorescences are also damaged.

Control methods

- Keep them away from plantations using air rifles and firecrackers. Apart from this, research is being carried out by the Coconut Research Institute to drive away monkeys using high-frequency sound waves.
- A device producing high-frequency sound waves is under testing.

Diseases in Coconut Plantations and their Management

1. Bud Rot Disease

- A fatal disease caused by the fungus *Phytophthora palmivora*.
- High humidity, stagnation of water, and gloomy weather conditions are favorable for growth of the fungus.
- Hence, the disease is common in rubber lands, river valleys and water-stagnating areas.
- This fungus infects coconut palms as well as rubber, cocoa, date palm and palmyra palm in the wet zone and dry zone.

Symptoms

- First, the bud turns pale and shows a dried-up appearance.
- Gradually the bud turns brown and dries up. Later, the base of the bud rots.
- Dry leaves fall off.
- The rest of the healthy leaves are green, and later they also dry out.
- Coconut inflorescences dry and nuts fall.
- Mature parts become soft and watery.
- At the end, a stem without a crown remains in the field.

Control Methods

- As spores of the fungus are spread by water, wind and insects, cut dead palm and burn the crown.
- If the disease is detected at an early stage, cut off the rotten parts.
- Make a solution with fungicide containing Mancozeb 64% + Metalaxyl 8% (w/w) WP by dissolving 4-5 g in 1 L of water, or 4 ml of a fungicide containing Tebuconazole 250 g/l EW. These chemicals can be used separately or in mixtures.
- Thoroughly wet the bud region with one of the above solutions.
- As a prophylactic measure, the disease can be prevented by thoroughly wetting the bud region of the palms around the diseased palms once every two weeks using Bordeaux mixture, or a fungicide containing Mancozeb 64% + Metalaxyl 8% (w/w) WP or a solution containing Tebuconazole 250 g/l EW.



Bud rot damage in mature coconut

2. Stem Bleeding Disease

- Caused by the fungus *Ceratocystis paradoxa*.
- This fungus is a weak parasite and enters the stem only through wounds/ cracks on the stem.
- This disease is usually common in immature palms.

Symptoms

- Symptoms include longitudinal fissures in the bark and the oozing of a red-dish-brown fluid.
- Later, the fluid dries to black on the stem surface.
- The bark of the stem turns yellow at the beginning due to internal tissue decay, and later turns blackish brown,
- leaving only fibrous fragments.

Control Methods

- Remove the diseased tissue with a piece of healthy tissue using a sharp tool and burn the removed diseased parts.
- Apply a 1% copper fungicide solution such as Bordeaux mixture or Dithiocarbamate.
- Apply 4 ml of fungicide containing Tebuconazole 250 g/l EW mixed with 1 L of water.
- After 10 days, apply coal tar/burnt engine oil only if there is no new oozing.
- If the wound has extended into the stem, close the hole with a cement-sand mixture after applying the fungicide.
- To prevent the fungus from entering through wounds on the stem, apply coal tar/burnt engine oil to the wounds.
- Avoid making any injury to the stem.



Symptom of Stem bleeding disease

3. Coconut Leaf Blight Disease

- A common disease in all coconut-growing areas.
- It is caused by 2 species of fungi called *Pestalotiopsis palmarum* and *Bipolaris incurvata*.
- A coconut palm of any age is susceptible to the disease.
- Although the coconut palm does not die from this disease, the palm's health declines.
- Susceptibility of the coconut palms to this disease is increased with high humidity, poor drainage, dry weather conditions, potassium deficiency and excessive nitrogen fertilizer applications.

Symptoms

- Small yellow spots on the coconut leaves gradually enlarge, turning brown and gray in the center of the spot.
- Several adjacent spots merge to form large brown areas.
- Lower fronds show symptoms first. In an epidemic situation, the spots spread to the upper fronds.
- When the disease is severe, the leaves dry up as if they are burnt.
- Sometimes this fungus infects the mid rib of the coconut fronds where the inner tissues of the fronds become discolored and dried.



Symptoms of coconut leaf blight disease

Control method

- To prevent disease,
 - Adopt proper management practices in the plantation such as:
 - Improvement of drainage.
 - Irrigation during dry periods.
 - Remove excessive shading.
 - Application of fertilizers at the right time and in the right amount.

- If the disease is present in a field where fertilizer is applied, Apply the following additional potassium fertilizer every 6 months until symptoms disappear.

	Muriate of Potash (MOP) (g)	Ash from the Kitchen (kg)
Per young plant	250	5
Per Adult Palm	500	10

- Spraying a solution of 4 ml of a fungicide containing Tebuconazole 250 g/l EW mixed in 1 L of water during an epidemic situation in nurseries.
- If the disease is epidemic in mature palms, spray a fungicide solution containing Tebuconazole 250 g/l EW on to fronds.
- If the fronds at the lower level of the palm are affected by this disease and gets dried, cut off diseased fronds or parts of fronds with the disease and spray a fungicidal solution.

4. Collar Rot Disease

- Seedlings in the nursery and those newly planted in the field are susceptible to the disease.
- The disease is caused by soil bacteria that enters the seedling through the growing tip.
- Black beetle damage and excessive moisture cause coconut seedlings to become susceptible to this disease.

Symptoms

- The bud of the coconut palm becomes withered, discoloured and dried up.
- Rotting of parts at the base of the bud and crown.
- The bud detaches easily from the seedling when pulled and the rotting parts produce a foul smell.

Control Methods

- Remove mulch during rains to remove excess moisture, stop irrigation and remove excess shade.
- Plant seedlings to expose the bud region to sunlight without burying it too deep in the soil.
- Remove and burn diseased seedlings.
- Improve drainage.

5. Ganoderma Root and Bole Rot Disease

- This disease was first reported from the Hambantota district in the Southern Province.
- It is caused by the fungus called *Ganoderma boninense*.
- The disease is common in palms older than 15 years.
- The spread of the disease is very slow, and it causes rotting of the roots and base of the stem.
- The palm dies 3–6 years after the onset of disease.
- The disease is primarily spread by physical contact of diseased roots with healthy roots.
- In addition to coconuts, this fungus infects arecanut palm, palmyra and other palm trees.

Symptoms

- First, a reddish-brown thick fluid oozes from the longitudinal cracks at the base of the stem.
- The secretion of this fluid gradually spreads up the stem.
- The yellowing of leaves that start from the lower fronds spreads to the upper fronds.
- Female flower and nut production is decreased.
- Shortening of fronds and thinning of the stem are observed at the advanced stage of the disease.
- Root and stem decay spreads up to the crown.
- At the base of the exposed roots and stem, bracts of the fungus with a reddish-brown upper surface and a lower whitish surface with small pores are formed.
- Eventually, the tree collapses at the base or dies as if in a drought.



Symptoms of Ganoderma Root and Bole Rot Disease

Control Methods

- To prevent the disease, uproot diseased palms with the roots and burn the base of the stem completely.
- To prevent the roots of the diseased palm from touching the roots of healthy trees, cut a 1 m deep and 30 cm wide trench around the diseased palms at 4 m from the base of the palm.
- Increase the activity of beneficial soil organisms and reduce the activity of parasitic fungi by applying organic manure in the palm basin.
- Fertilizer application, irrigation during the dry season and improvement of drainage conditions during the rainy period can reduce the susceptibility of coconut palms to disease.
- Use 5 L of Bordeaux mixture in coconut pits at planting coconut seedlings in previously diseased areas.

6. Leaf Die-Back Disease

- The disease is caused by the fungus *Curvularia lunata*.
- Infect seedlings in the nursery as well as newly planted palms in the field.
- Weakening coconut palms increases the susceptibility to disease.

Symptoms

- Drying of fronds/leaves of coconut seedlings from the tip of the fronds.
- This drying starts from older leaves and spreads to younger leaves.
- Such leaves become weak and may die if left untreated.

Control Methods

- To prevent the disease, adopt measures such as:
 - o Regular weed control
 - o Improve drainage
 - o Irrigate during dry periods
 - o Apply a balanced fertilizer mix
 - o If the disease spreads in epidemic conditions, dissolve 4 ml of 1% copper fungicide / Tebuconazole 250 g/l EW fungicide in 1 L of water and spray onto the leaves until the surfaces are thoroughly wet.
 - o Irrigate without wetting the leaves

7. Weligama Coconut Leaf Wilt Disease (WCLWD)

- Coconut palms of all ages are susceptible to the disease.
- The disease is caused by a pathogen called Phytoplasma and is spread by several species of sap-sucking insects.
- As the phytoplasma remains in the palm until the diseased palm is dead, it is capable of spreading the disease during the entire period of its presence in the field.

Symptoms

- The angular nature of the coconut leaves is lost, and the leaf blade becomes flat, and the leaves fold diagonally like the ribs of humans.
- 10-12 coconut fronds from the lower to the middle whorl turn intense yellow.
- This yellowing occurs unevenly. Not all coconut leaflets on the same frond are uniformly yellow, and some leaflets appear healthy green.
- Drying of yellowed leaves starting from the edge and yellowing spreading inwards.
- Cracking of the edges of the coconut fronds in the middle and upper layers of some palms.
- Roots appear as nodules due to rotting of the internal parts behind the root sheath of newly formed roots.
- Due to the disease, the yield gradually declines, and the stem becomes thinner.

Control Methods

- Although the affected palm does not die due to the disease, the vigor and productivity of the palm is gradually decreased.
- Palms infected by the Weligama coconut leaf wilt disease are secondarily infected by the leaf rot disease.
- So far, no treatment has been found worldwide to control the diseases caused by Phytoplasma.
- The spread of the disease can be prevented only by destroying the diseased trees by:
 - o Cutting diseased palms and destroying their crowns.
 - o Avoid transporting palms or live plant parts from diseased to healthy areas.
 - o Avoiding the planting of seedlings produced in the diseased area.
 - o Planting only the coconut varieties that are resistant to the disease in the affected area. e.g. green dwarf and resistant seedlings specifically produced by the Coconut Research Institute for this purpose



Coconut palm affected by
Weligama Coconut Leaf Wilt disease

8. Leaf Rot Disease

- Palms weakened by Weligama coconut leaf wilt disease are secondarily infected.
- The disease is caused by several fungi infecting the flag leaf.

Symptoms

- First, small, brown and water-soaked spots of various shapes appear on the white young leaves of the bud (flag leaf).
- As the spots enlarge, they connect to form large, rotten areas.
- Small brown spots on leaflets.
- Although the rot is limited to the young leaves, in some cases, it can spread to the growing tip.
- When the young leaves unfold, the rotten parts appear black.
- The tips of leaves and rachis turn black and shrink.
- As the fronds mature, the rotting stops and the centre of the leaf rachis appears as a fishing rod when unfolded.
- When the disease is severe, the fronds fall off, the stem becomes thin, and the emergence of inflorescence and nuts is reduced.



Coconut palm affected by leaf rot disease

Control Methods

- Cutting down diseased palms and burning the crown part.



Weed Management in Coconut Plantation

- Weed control is very important in a coconut plantation.
- Although the damage caused by weeds to the coconut palm is low compared to other crops, weeds can cause economic damage in young and bearing plantations.
- Weeds compete with the crop for soil nutrients, soil moisture, sunlight and space, thereby reducing the growth and yield of coconut palms.
- The high weed population in coconut fields creates obstacles to all management practices, including fertilizer application, coconut harvesting and collection of nuts.
- In this case, an additional expense has to be incurred for weed control, because of which the cost of production of coconuts increases, and ultimately the profit received by growers decreases.
- In the process of weed control, complete weed eradication from coconut fields is not expected, and it is sufficient to control the weed population to a level that does not cause economic damage to the coconut plantation.
- Commonly used weed control methods are:
 1. Mechanical
 2. Agronomic
 3. Biological
 4. Chemical, and
 5. Integrated weed management

1. Mechanical methods

- The use of mechanical methods such as hand pulling, manual slashing and hoeing is cost-effective for small-scale plantations.
- For large-scale plantations, weed control can be successfully achieved by tractor-mounted slashers and by harrowing the land.
- Removing weeds with grass cutters is not complete weed control; it only controls the weeds temporarily.
- Mechanical weed control should be done before flowering.
- Broad-leaved weeds such as *Chromolaena odorata* (Siam weed or Podisinghamaran), *Lantana camara* (Lantana or Gandapana), and grasses like *Imperata cylindrica* (Cogon Grass or Illuk), *Megathyrsus maximus* (Guinea grass), *Pennisetum polystachion* (Mission grass or Maana), *Panicum repens* (Torpedo grass or Etor), can be successfully controlled by uprooting them along with the basal part of the plant and root system. Removing the above-ground growth of such weeds will not be sufficient to control them without removing the below-ground parts, as the plant's basal sections and root system continue to sprout from the soil.

2. Agronomic methods

- i. Cover cropping
- ii. Intercropping
- iii. Livestock integration

i. Cover cropping

- Cover crops are essentially surface feeders and generally do not compete for moisture with relatively deep-rooted plantation crops.
- Cover crops reduce soil erosion by preventing surface runoff and falling of raindrops directly onto the ground.
- Provides a large amount of leaf litter and organic matter to the soil.
- Improves soil structure and water absorption capacity and conserves moisture.
- Legume cover crops provide nitrogen nutrients to the soil.
- Vine Cover Crops – Pueraria, Calopogonium, Ciratro and Centrosema
- Bush Cover Crops – Gliricidia, Acacia, Ipil Ipil (*Leucaena leucocephala*)
- Leaves of the above plants can be used as mulch to cover the manure circle of coconut palms.

ii. Intercropping

- When the age of the coconut plantation is less than 5 years and more than 20 years, sunlight reaches the ground adequately, and the possibility of weed growth increases.
- Therefore, the growth of weeds can be minimized by planting suitable intercrops in the plantation during these stages.
- Cultivation of crops such as black pepper, coffee, cocoa, pineapple, and banana leads to suppression of weeds, reduction of soil erosion and increase of soil fertility.

iii. Livestock integration

- One animal per 3 - 4 acres is recommended when grazing by buffalo and cattle is practiced.
- Six animals per acre are recommended for goats and sheep.
- *Imperata cylindrica* and *Megathyrsus maximus* can be controlled successfully by buffaloes.
- Since goats damage young coconut palms, they should be reared only in mature coconut plantations.
- However, sheep can be reared in young coconut plantations.

3. Biological methods

- Insects are used to control weeds.
- Although this approach is not widely used presently, this method has been used successfully to control several weeds.
- E.g. A caterpillar called *Amelo insulata* can be used to control *Chromolaena odorata*. This caterpillar eats the leaves of the plant and gradually the plant becomes weak and dies. This is cost-effective and minimizes environmental damage.

4. Chemical methods

- In this method, weeds are controlled by herbicides.
- Soil moisture and stage of weed growth should be considered when applying herbicides to get the most effective results.
- Overuse of herbicides has a negative impact on the environment.
- For optimal results, remove the mature weeds from the surface, let them remain for two to four weeks, and then apply herbicide to the emerging shoots as soon as they emerge from the plant parts.

Recommendation for controlling *Panicum maximum*

i. For mature stands

Mix 100 ml of Glufosinate ammonium with 16 L of water (using a knapsack sprayer) and apply 8 - 10 tanks per acre.

ii. Immature stands before flowering (after slashing or harrowing)

Mix 80 ml of Glufosinate ammonium with 16 L of water (using a knapsack sprayer) and apply 8 - 10 tanks per acre.

5. Integrated weed management

- Integrated weed control refers to the use of multiple methods instead of using one approach.
e.g.,
Mulching the manure circle and growing intercrops in the remaining land.
- Applying herbicides in the manure circle and removal of weeds in the remaining area using a mechanical methods.



Weeding using a grass cutter

Maximizing Profit Through Intercropping, Animal Husbandry, and Integrated Farming

In a coconut plantation, typically three-fourths of the land remains unutilized, which can be used to grow other intercrops in addition to coconut palms. Intercropping coconut with other crops will create a mini-forest atmosphere and increase biodiversity.

Principles of intercropping

- Income generation from intercrops
- Efficient utilization of resources (soil, soil water, sunlight and plant nutrients)
- Improves soil water infiltration and aeration
- Reduces soil erosion and increases soil fertility
- Increases coconut production
- Enables weed control
- Reduces the risk of dependence on only one crop (coconut) for income
- Enhance food and nutritional security
- Create more employment opportunities

Considerations for growing intercrops

1. Application of fertilizer

- Recommended amounts of fertilizers should be applied separately for coconut and intercrops, to avoid competition for nutrients.

2. Light

- In a coconut plantation, it is advisable to grow light-loving plants until the age of 5 and after 20 years.

3. Soil moisture

- The competition between the coconut palm and intercrops for moisture can be reduced by planting intercrops more than 1.8 meters (6 feet) away from the coconut palm.
- Only intercrops that can survive under such conditions should be chosen in situations where soil moisture is a limiting factor (such as sandy soil).

4. Location of land

5. Soil type

- Soil erosion should be reduced in sloping coconut lands.
- Different types of soils are found in different climatic zones.
- The growth of coconut palms also changes depending on soil types.
- S1 (Soil suitability class) soils give higher yields than other soil suitability classes.
- It is challenging to grow intercrops in S1 soil because of the limited light in those areas.
- Lands with shallow, gravelly soils have low coconut productivity and are suitable for intercropping.

6. Market demand

- When choosing intercrops, the existing marketing facilities in the area should also be taken into consideration.
- By growing several intercrops, one can cope with the fluctuation in prices.

7. Social factors and other facilities

- Factors such as the availability of fertilizer, the willingness/knowledge of growers, and government subsidies should also be considered before introducing intercrops.

Types of intercrops that can be grown

1. Annuals that are grown for less than a year

- Legumes - Cowpeas, Green Beans, Soybeans, Long Bean
- Yams - Sweet Potato, Cassava, Kiri Ala, Wel Ala
- Spices - Chillies, Ginger, Turmeric
- Different vegetables

2. Medium-duration crops last for 4 - 5 years

- Fruits - Banana, Pineapple, Orange, Papaya
- Minor Export Crops - Betel
- Medicinal Herbs - Aloe Vera, Thippili, Inguru Piyali

3. Long-term crops lasting for more than 10 years

- Cocoa, Coffee, Black Pepper, Cloves, Cinnamon, Rambutan, Lemon, Mandarin, Nutmeg, Tea and Fodder Grasses are suitable.
- In addition, crops like Cashew, Lime and Mango are suitable for intermediate and dry zones.

Intercrops recommended for coconut cultivation

Several intercrops can be grown along with coconut palms to enable the growers to get additional income to balance the expenditure and income from the plantation.

Coconut with Black Pepper

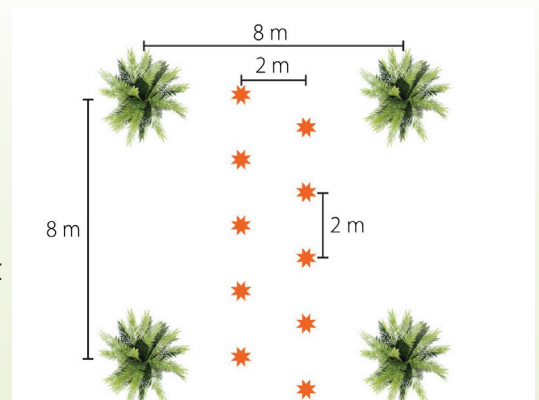
Black Pepper is the most widely used spice in the world and is known as 'King of the Spices'. Black Pepper is native to South Asia, and historical records reveal that it originated in South India. This is one of the most profitable intercrops for coconut fields in the wet zone. Since black pepper is a perennial spice with a life span of over 20 years, it could be easily maintained after establishing in the field.



Varieties - Panniyur-1, Kuchin, Dingi Rala, Kohukumbure Rala and Bootawe Rala.

Spacing - Inter row 2 m x Intra row 2 m

Planting Methods - Coconut plantations aged 15-50 years are suitable for intercropping black pepper. They are mainly grown in double row systems. In this system, two rows of black pepper are planted 2.4 m apart between two rows of coconut. The black pepper plants should be spaced at 2 m within a row and arranged in a triangular pattern.



Training and Pruning Black Pepper

When 8 -10 leaves have emerged, all leaves should be removed except the lowermost three leaves. After one week, prune the defoliated portion of the vine. This will encourage the growth of 2 or 3 shoots. Again, when these shoots produce 8-10 leaves, repeat the removal of leaves and pruning as before. Finally, each vine will have 7-9 branches. These branches should be tied to the support using twine or banana strands. The apical buds should be removed when the vines grow about 3 m to restrict vertical growth. The runners that grow along the ground should be removed at the time of fertilizer application.

Yield - A well-managed black pepper vine at the age of 5 years will yield 2-3 kg of processed black pepper annually.

Coconut with Pineapple

Pineapple is one of the major commercial fruits grown in Sri Lanka, with a high demand in the export market. Second only to bananas, pineapple is a very popular, profitable and tasty fruit crop intercropped in coconut fields. The favourable climatic conditions, high consumer demand, supplementary income and availability of coconut lands with suitable soils in the wet and intermediate zones are the contributory factors for pineapple being widely intercropped.



Varieties - Mauritius, Kew

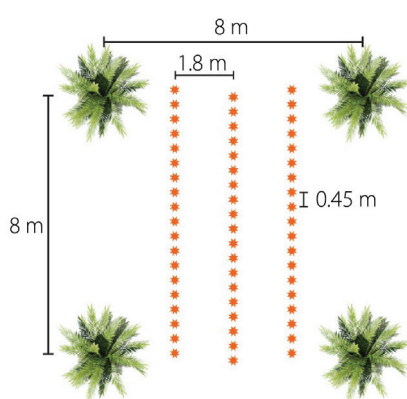
Spacing -

Single row method -

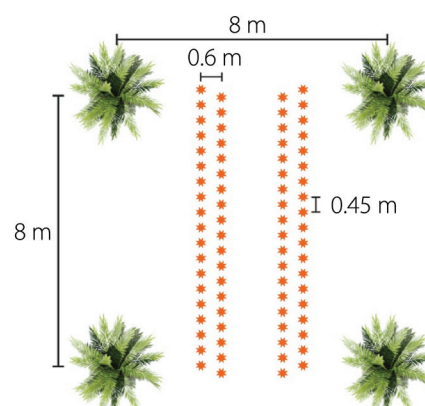
Inter row 1.8 m x Intra row
0.45 m

Double row method -

Inter row 0.6 m x Intra row
0.45 m



Single Row



Double Row

Planting Methods - Coconut plantations younger than 5 years and older than 30 years are suitable. The main planting methods are single row and double row. Prepare trenches 20 cm wide and 20 cm deep. In sloppy lands, prepare a contour system and in flat lands, prepare rows in east and west directions. Keep the suckers in the trenches and add the soil to cover 10-15 cm of the sucker base.

Artificial flowering - To avoid uneven natural flowering, artificial flower induction is common in commercial practices. Naphthalene, Acetic acid (NAA - plantfix) or Ethephon (ethylene-releasing chemical) can be used. 35-40 leaf stage is the most critical stage for flower induction. Do not use chemical fertilizer and weedicides one month before hormone application.

Harvesting - Well-matured fruits with 25% colour change from the fruit base are suitable for harvesting. Fruits should be harvested with 10-15 cm of stem. A fully developed fruit of the Mauritius variety weighs about 2 kg, while it is 4-5 kg in the Kew variety.

Coconut with Banana

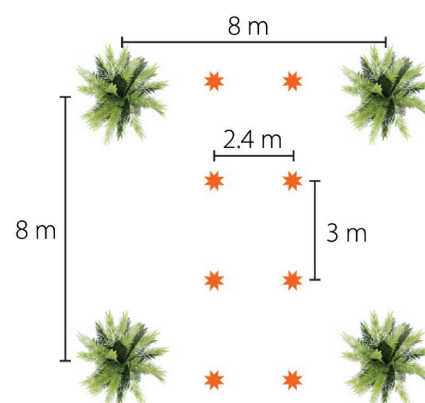
Banana is a very popular fruit due to their low price and high nutritional value. It is consumed in fresh or cooked forms, both as ripe and raw fruit. Intercropping bananas in coconut fields is popular among growers compared to other intercrops. The reasons for its popularity are the availability of planting materials, fast growth, an available market, and high returns.



Varieties - Nadee (Ambul-AAB), Kandula (Dual Purpose-AABB), Pulathisi (Cooking Banana- ABBB), Prasad (Ash Plantain-ABB), Agra (Kolikuttu-AAB), Parakum (Seeni Kesel-ABB), Millewa Suwandel (Suwandel-AAB), Gannoruwa Seeni1 (Seeni Kesel-ABB), Gannoruwa Red Banana (Rath Kesael-AAB).

Spacing - Inter row 2.4 m x Intra row 3 m

Planting Methods - Coconut plantations older than 30 years are suitable for intercropping banana." Planting holes of 60 cm x 60 cm x 60 cm are sufficient. Those holes should be filled with a mixture of top soil and organic manure. In addition to that, 500g of Rock Phosphate and 1 kg of dolomite should be added to the planting hole before planting.



Harvesting - Generally, flowering is initiated 10 - 12 months after planting. However, the time taken to maturity depends on the cultivar. Ambul types take 13 weeks while Kolikuttu takes 16 weeks.

Coconut with Cinnamon

True cinnamon is native to Sri Lanka. Cinnamon is originally grown wild in the up country of Sri Lanka. The history of cinnamon dates back about 2800 B.C, where it can be found referenced as "kwai" in Chinese writings. In Sri Lanka, cinnamon seems to have originated in the up country where seven wild species of cinnamon occur in Kandy, Matale, Belihull Oya, Haputale, Horton Plains and the Sinharaja forest range. Presently cultivation is concentrated along the coastal belt from Negambo to Matara. It is also grown in Kalutara and Ratnapura as well.



(a) Cinnamon under Coconut

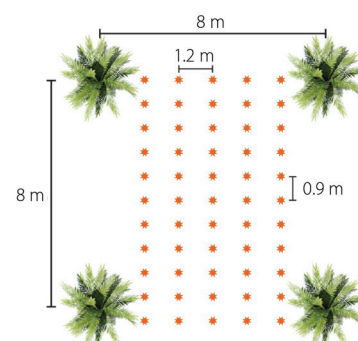


(b) Cinnamon processing

Varieties - “Pani-Miris Kurundu” is the best with sweet-pungent taste and “Miris Kurundu”, “Sevel Kurundu” and “Thiththa Kurundu” are the others. Currently ten cinnamon accessions have been identified based on yield and quality performances and best two lines, named as “Sri Vijaya” and “Sri Gamunu”, have been released. Other selections are under evaluation in different agro climatic zones.

Spacing - Inter row 1.2 m x Intra row 0.9 m

Planting Methods - Coconut plantations of 25 - 45 years are suitable. Well-ripened seeds are selected, thoroughly washed to remove pericarp and planted in 12.5 x 20 cm poly bags filled with equal parts of top soil, cow dung, sand and coir dust. Five to eight seeds are planted in a bag, but thinning out is done to keep 4-5 vigorous plants after about two months. Planting is done with the onset of monsoon rains. Healthy, disease-free, four-month-old seedlings are planted in pits of 30 cm x 30 cm x 30 cm. Planting holes are filled with topsoil and cow dung or compost and one bag with 4-5 seedlings is planted in a pit. Vegetative propagation through stem cuttings is also possible.



Pruning Methods - After harvesting the sticks, many branches are produced from the plant at the same time. All the weak branches should be pruned, leaving only the healthy and erect branches. This practice, which is called selective pruning (Nuhuru paheima), increases the cinnamon yield.

Harvesting - The First harvest of cinnamon can be taken after three years of planting, and two harvests can be taken per year. Harvesting is done when the bark colour of the stem turns brown and the stick diameter is about 3-5 cm. Branches and leaves are removed from the harvested sticks before peeling, and harvested stems should be peeled on the same day. Approximately 200-250 kg of cinnamon can be harvested annually from one acre of coconut plantation intercropped with cinnamon.

Coconut with Ginger

Ginger is a popular spice crop that can be easily intercropped in coconut fields. Local ginger production mainly arises from those grown in coconut lands. Several local and introduced varieties are grown in Sri Lanka on a large scale. Ginger is grown all over the country, but the wet and intermediate zones are major growing areas. Kurunegala, Kandy, Gampaha, Colombo and Kegalle Districts are the main ginger-growing areas. Ginger is largely grown as an intercrop with coconut and as a home garden crop.



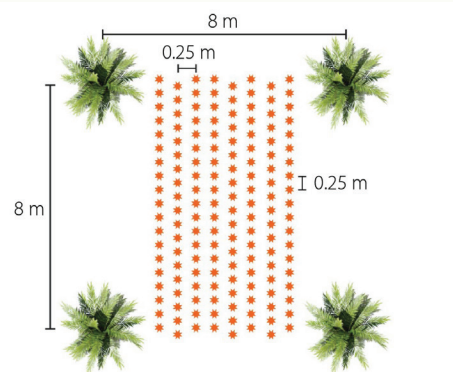
Varieties

Local ginger - Rhizomes are small, and the fibrous flesh is somewhat ash white in color. Pungency and aroma are comparatively higher than other ginger varieties. Largely used for beverages.

Chinese ginger - Rhizomes are large with watery flesh. Flesh is pale yellow in color. The pungency and aroma are low. Largely used for pickles.

Rangoon ginger - Rhizomes are medium in size and with well-developed finger rhizomes.

Planting Methods - Coconut plantations older than 20 years are suitable. Field should be ploughed up to 35-40 cm in depth and tilling done. Beds can be 105 cm in width and length, which vary with the space available. However, when intercropped with coconut width and length can be changed according to the available space. Height of the bed is 22 cm and a drain, with 45 cm in depth, should be prepared between beds. A spacing of 0.25 m x 0.25 m is suitable. Approximately 600 - 800 kg of seed ginger is required per acre of coconut plantation.



Coconut with Turmeric

Turmeric has been used in India for over 5,000 years. Initially, it was cultivated as a dye due to its vibrant yellow color, which made it an excellent coloring agent. Over time, its versatile applications became apparent, leading to its use in cosmetics and beauty treatments, and eventually as a medicinal remedy. Turmeric is grown in wet and intermediate zones of Sri Lanka as a monocrop and an intercrop under coconut. Major growing districts are Kurunegala, Gampaha, Kalutara, Kandy, Matale

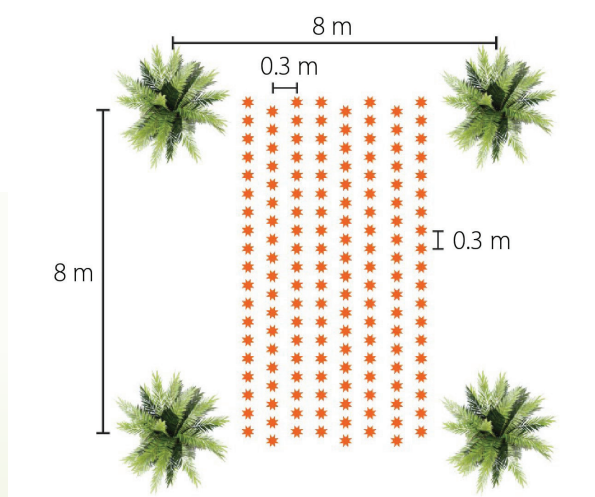


and Ampara.

Varieties – Though there are a number of locally grown varieties, they are not identified. Imported varieties namely, Gunter, Puna and Madurasi Majal are available, and they have been mixed with local varieties.

Turmeric rhizomes are classified into two types: mother rhizomes and finger rhizomes. Among these, mature finger rhizomes are considered the most suitable planting material. Each rhizome piece should weigh between 30-50 g and have 1-2 buds for optimal growth.

Planting Methods – Coconut plantations of more than 20 years old are suitable for intercropping with turmeric. The field should be ploughed up to 35-40 cm in depth and tilling done. Beds of 120 cm in width and 3.0 m length should be prepared. However, when intercropped with coconut, width and length can be changed according to the available space. The height of the bed is about 20 cm and drains, with 30 cm in depth, should be prepared in between beds. A spacing of 0.3 m x 0.3 m is suitable. Approximately 600 - 800 kg of seed turmeric is required for one-acre coconut plantation.



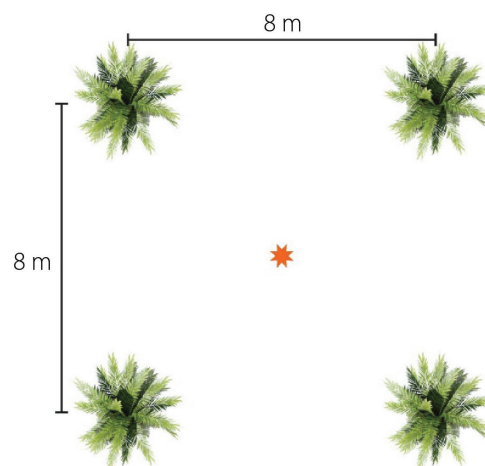
Coconut with Cashew

Cashew is very popular cash crop due to its drought tolerance ability and low cost of production. Cashew does not require fertile soil and marginal lands are suitable. Hence, there is a greater potential in further expanding cashew cultivation in Sri Lanka. Currently, it is cultivated in Puttalam, Wanathavillu, Mannar, and Batticaloa.



Varieties – The varieties in “WUCC (Wayamba University – Cashew Cooperation)” group developed by Sri Lanka Cashew Corporation in collaboration with Wayamba University of Sri Lanka (WUCC 05, WUCC 08, WUCC 09, WUCC 13, WUCC 19 and WUCC 21)

Spacing – One plant per coconut square

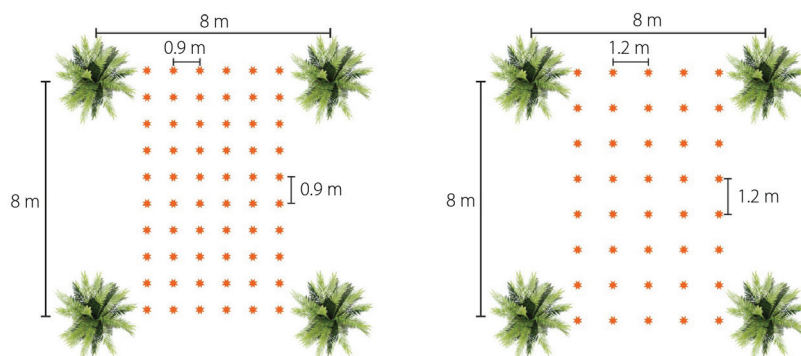


Planting Methods – Select fields where coconut is not growing well (marginal lands). Coconut plantations with wider spacing are more suitable for intercropping with cashew. Planting in a pit of 60 cm x 60 cm x 60 cm dug in the center of the coconut avenue and filled with a layer of coconut husk and compost should be used for faster growth of cashew plants.

Harvesting – Generally, cashew flowers in January. The harvesting season in Gampaha area is February - May while it is March - June in Puttalam area. On an average, cashew plant produces about 5 kg of cashew annually. Management methods have to be intensified to increase up to 10 kg.

Coconut with Cassava

Cassava is an important root crop that provides food for over 500 million people in the world. It is a consistent and inexpensive food source, assuring global food security by providing the dietary carbohydrate requirements of people, especially in low-income countries. Coconut lands in the wet, intermediate and dry zones are suitable for cultivating cassava. It is commercially grown in Gampaha, Embilipitiya, and Moneragala areas, and large-scale growers cultivate continuously in the same land.



Varieties – MU-51, Kirikawadi, CAR 5555, Suranimala, Swarna, Shani, HORDI-MU-1, HORDI 06

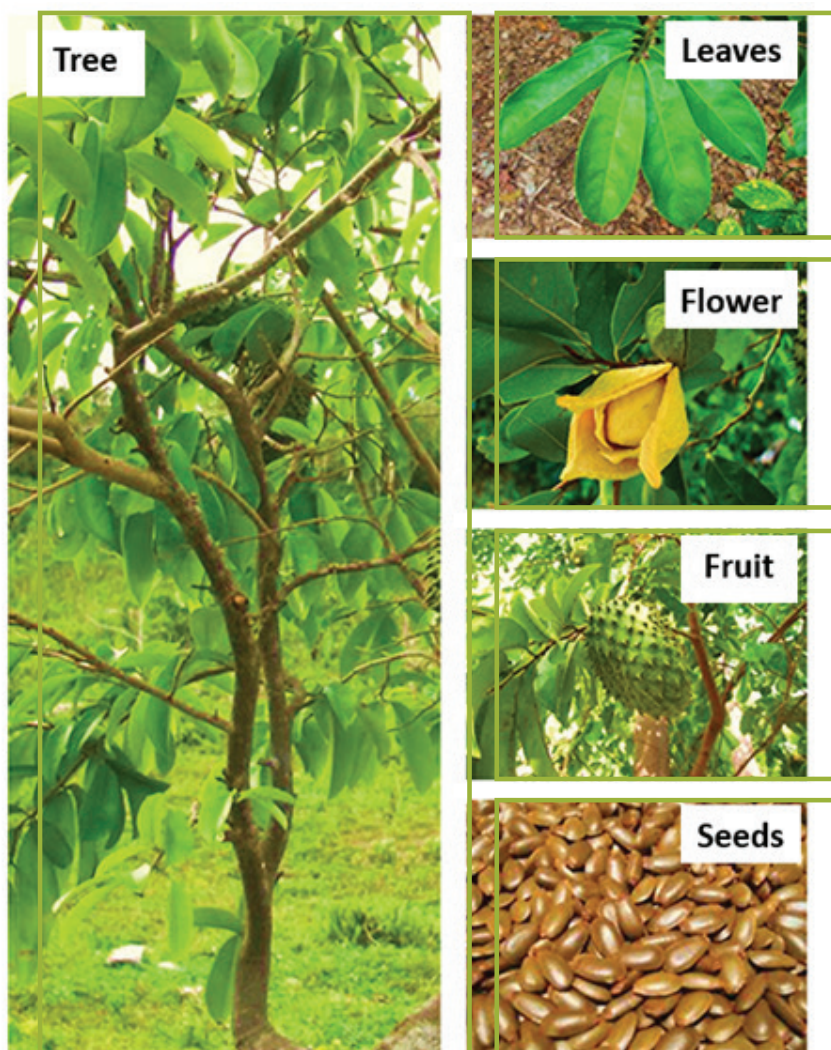
Spacing – Unbranching – Inter row 0.9 m x Intra row 0.9 m; Branching – Inter row 1.2 m x Intra row 1.2 m

Planting Methods – Coconut plantations with less than 5 years old plants and plantations older than 30 years are suitable. Stem cuttings of 10-15 cm are used as planting material. They should be planted before drying out to establish a vigorous cultivation. Planting should be done while soil is moist. The most popular method is planting stem cuttings at an angle of 45°.

Harvesting – The maturity period of the tubers varies from 4 to 9 months depending on the variety. Falling leaves, stem turning brown and sometimes splitting of soil are the signs of maturity of tubers. On average, a single cassava plant can produce a yield of 3-10 kg.

Coconut with Soursop (Annona)

Annona is a genus of tropical fruit trees in the Annonaceae family, comprising around 119 species. One notable species, *Annona muricata* L., is commonly known as Soursop in English-speaking regions and has various other common names. Soursop trees are widely distributed across tropical and frost-free subtropical areas, including the West Indies, the Americas, lowland regions of Africa, the Pacific Islands, and Southeast Asia. Despite their presence, the Soursop fruit and other tree parts are considered underutilized. The tree itself is upright with low-branching and typically grows to a height of 8 to 10 m.



Annona muricata L. Plant

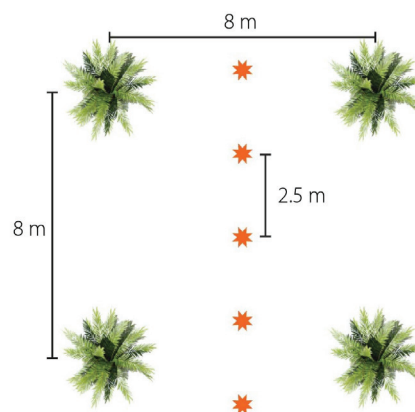
Varieties – Local varieties are more suitable.

Spacing – Plant at 2.5 m intervals using the single row method

Planting Methods – Pits of 60 cm x 60 cm x 60 cm should be prepared and filled with a mixture of top soil and cow dung/compost in equal quantity and after about a week, the selected healthy plants should be planted.

Pruning – Once the plant reaches a height of approximately 1 m, it should be pruned to a height of 80 - 90 cm. Allow only 3 - 4 branches to grow in different directions, ensuring a spacing of about one foot between them. Shape the canopy to allow sunlight to penetrate the center of the tree effectively. Prune branches at intervals of about 12.0 m to encourage secondary growth. Prune the dichotomous branches to maintain a spacing of approximately 30 cm between them. Similarly, trim the tertiary branches to ensure a spacing of about 20 cm. Shape the branches to spread evenly in all directions. Maintain the overall height of the tree at approximately 3 to 3.5 m.

Harvesting – The fruit is ready for harvest when the protrusions/thorns of the fruits are fully developed, the color changes from dark green to light yellow green, the protrusions/thorns begin to separate, and the surface of the fruit becomes smooth. Harvesting should be done before 10 a.m. or after 3 p.m., avoiding rainy days. The fruits typically ripen within 3-5 days after harvesting.



Coconut with Papaya

Papaya is a very popular and nutritious fruit. It bears a year-round offering for farmers with a quick income. Papaya is widely available in the market and highly favoured by consumers. Papaya can be grown as a fruit plant in all three agroclimatic zones. It yields high in dry zones if irrigated.

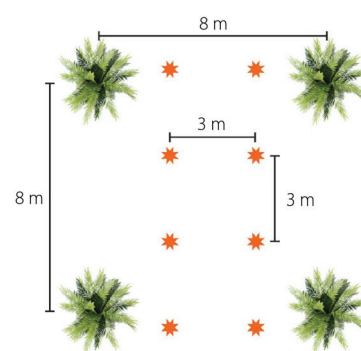
Varieties – Rathna, Red lady, Horana Papaya Hybrid 1

Spacing – Plant at 3 m intervals using the double row method

Planting Methods – New or replanted coconut stands having abundant sunlight are suitable. The dimensions of the planting hole are 60 cm x 60 cm in hard, gravelly soil and 30 cm x 30 cm x 30 cm in sandy soil.

The planting hole should be filled with compost, and plant two seedlings per pit. Unsuitable plants should be removed at the flowering stage.

Harvesting – It takes about 10-12 months to produce a significant yield. About 30-40 fruits can be harvested from a plant during a period of about 3 years.



Coconut with Mango

Mango is one of the most important and widely cultivated tropical fruits in the world. Often referred to as the “King of Fruits,” it belongs to the family Anacardiaceae and originated in the Indo-Burma region over 5,000 years ago. Rich in vitamins A and C, mango has established itself as a leading fruit crop worldwide. Mango is a drought-tolerant plant and has become popular as an intercrop in coconut plantations due to its low maintenance costs. Pest and disease damage is minimal, and the fruit maintains high market demand. Growers commonly practice cultivating mango trees in their home gardens as well.

Varieties – Mango varieties were introduced by the Department of Agriculture (DOA) according to their adaptability in different agro-ecological regions. The selection criteria of varieties were high yield, unique taste and long shelf life. Mango varieties recommended for different regions are as follows:

Wet zone – Horanahiru, Velleicolomban, Dampara, Giraamba

Intermediate zone – Karthakolomban, Velleicolomban, Malwana

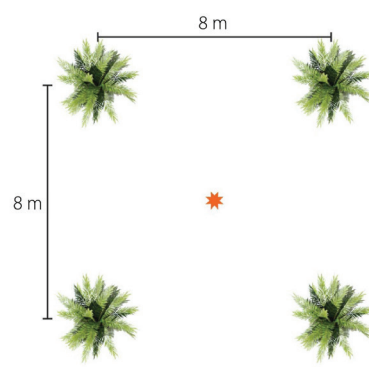
Dry zone – Karthakolomban, Vilard, TOM EJC, Malwana

Karthakolomban and TOM EJC, which have become popular recently, have high market demand. The fruits of TOM EJC are big, yellowish and very tasty. These seedlings have a high demand, and the grafted ones are more suitable for planting. Seedlings certified by the DOA should be obtained from approved nurseries of the DOA.

Spacing – One plant for a coconut square.

Land preparation– The planting hole should be prepared by digging a pit measuring 1m x 1m x 1m in the center of the coconut avenue. The hole should be filled with a mixture of coconut husks, cow dung, and compost.

Planting Materials– Budded or grafted plants are recommended. The plant should reflect the characters of its variety and label, which are certified by the DOA.



Training of mango tree: Allow the trees to grow upright for some time. When they reach about 50 cm in height, pinch the terminal bud to encourage lateral shoots to sprout. Once the shoots mature, pinch the terminal bud on each shoot again. Repeat this process until the end of the third cycle. Additionally, remove low hanging branches, twigs, and disorderly branches inside the tree to open up the canopy, as well as any diseased or dead branches.

Animal Husbandry

Poultry Integration

Poultry integration under coconut plantations demonstrates efficient land use through vertical integration of agricultural systems. The coconut canopy provides natural shelter for birds, protecting them from harsh weather conditions, predators, and excessive heat. Farmers can choose between raising layers for egg production or broilers for meat, utilizing the open spaces between coconut palms effectively. The system allows for free-range or semi-intensive management approaches, where birds actively help control weeds and insects while naturally fertilizing the soil with their droppings.



Integration of poultry with coconut

Mobile chicken coops can be strategically rotated throughout the plantation to distribute manure evenly and prevent soil degradation in any area. Birds benefit significantly from the cooler microclimate under coconuts, which reduces heat stress and improves overall productivity. The integration reduces feed costs as birds forage on insects, fallen seeds, and grass available in the plantation. Proper disease management and biosecurity measures are essential for successful integration, including regular vaccination schedules and health monitoring. The system provides regular income through selling eggs or meat while improving soil fertility through natural fertilization. Farmers must ensure adequate housing facilities, proper ventilation, and protection of birds from predators. Regular cleaning and maintenance of housing units to help prevent disease outbreaks. Integration creates a sustainable farming system with multiple income streams and reduced input costs.

Goat Integration

Goat integration in coconut plantations exemplifies sustainable integration of livestock with tree crops. Goats efficiently consume various grasses, legumes, and broadleaved weed plants growing under coconuts, effectively controlling weeds, and minimizing the need for chemical herbicides. The system requires careful planning with proper fencing and rotational grazing strategies to prevent damage to young coconut palms and ensure sustainable fodder availability. Goats provide multiple income sources through meat production, milk, and selling young goats, while their manure significantly enriches soil fertility.



Integration of goat with coconut

Their droppings can be collected for composting or direct application as organic manure. The coconut canopy provides essential partial shade, reducing heat stress on the animals and creating a comfortable environment for grazing. Farmers need to implement comprehensive management practices, including adequate supplementary feeding during lean periods, proper housing facilities for protection from adverse weather, and regular monitoring of healthcare. Integration helps reduce labor costs for weed management while generating consistent additional income. Dwarf or local goat breeds are typically preferred for easier management under coconuts due to their adaptability and lower feed requirements. Regular health checks, deworming, and vaccination programs ensure healthy stock maintenance. Proper breeding management helps maintain herd quality and productivity. The system creates a sustainable cycle where goats benefit from the plantation environment while contributing to its maintenance and improving productivity.

Cattle Integration

Cattle integration in coconut plantations represents a comprehensive approach to combining dairy or beef production with plantation maintenance. Cattle effectively graze on understory vegetation, controlling weeds while converting otherwise unused biomass into valuable products, including milk, meat, and manure. The system requires careful management strategies to prevent soil compaction and damage to coconut roots, particularly during wet seasons. Cattle provide a regular income through milk production and selling calves, while their dung significantly improves soil fertility and can also be utilized for biogas production. The coconut canopy helps moderate temperature extremes, creating a more comfortable environment for cattle and reducing heat stress. Farmers must ensure adequate availability of water throughout the plantation, proper housing facilities for shelter during adverse weather, and comprehensive healthcare facilities including regular veterinary care. 'Cut and carry' feeding systems may need to be implemented during wet seasons to prevent soil damage from excessive trampling. The integration significantly reduces chemical inputs for weed control while providing a steady income through dairy or beef production. Proper breeding management helps maintain herd quality and productivity. Regular health monitoring, vaccination programs, and parasite control measures are essential for maintaining healthy cattle. The system creates a sustainable agricultural model where cattle contribute to plantation maintenance while generating additional income streams.



Integration of cattle with coconut

Sheep Integration

Sheep integration under coconuts creates an efficient biological weed control system combined with meat production. Sheep demonstrate a distinct preference for grazing close to the ground, effectively maintaining low vegetation without causing damage to coconut root systems. The integration works optimally with carefully calculated stocking rates and well-planned rotational grazing systems to prevent overgrazing and ensure sustainable fodder availability. Sheep provide multiple income sources through meat production, wool, and selling lambs, while their manure contributes significantly to soil fertility improvement.



Integration of sheep with coconut

The coconut canopy offers essential protection from extreme weather conditions, creating a more comfortable environment for the animals.

Farmers need to implement comprehensive management systems, including proper fencing for controlled grazing, regular healthcare monitoring, and supplementary feeding programs when necessary. Local or crossbred sheep adapted to the local climate conditions are preferred for better survival and productivity. The integration significantly reduces labour costs for weed management while providing a regular income through sheep products. Special attention must be given to parasite management due to the humid conditions typically under coconuts. Regular health checks, deworming programs, and vaccination schedules ensure healthy flock maintenance. Proper breeding management helps maintain flock quality and productivity. The system creates a sustainable agricultural model where sheep contribute to plantation maintenance while generating consistent additional income.

Fish Integration

Fish integrated within coconut plantations maximize water resource utilization while creating additional income streams. Farmers can construct specialized ponds between coconut rows or utilize existing water bodies, creating a productive aquaculture system. The integration can support various fish species depending on water quality parameters, depth, and local climate conditions. The partial shade provided by coconut palms helps to maintain optimal water temperature and reduces excessive algal growth, creating better conditions for fish cultivation. Fish feed on natural pond organisms, while their waste provides valuable nutrients for coconut palms through seepage and when pond water is used for irrigation.



Integration of fish with coconut

Integration requires comprehensive pond management strategies, including regular water quality monitoring, appropriate stocking densities, and proper feed management. Farmers can implement polyculture systems, culturing multiple fish species together for better resource utilization and increased productivity. Regular monitoring of water parameters including pH, dissolved oxygen, and ammonia levels ensures optimal growing conditions. Proper disease prevention measures and regular health monitoring help maintain healthy fish populations. The system provides regular income through fish sales while creating a more diverse and resilient farming system. Integration of duck farming with fish ponds can further enhance productivity. Periodic pond maintenance, including bottom cleaning and water quality management, ensures sustainable production. The system creates an efficient water-use model while generating additional income through fish production.

Integrated Farming in Coconut

Coconut-based integrated farming systems represent a highly efficient and sustainable agricultural approach that maximizes land use and resources effectively through the strategic combination of multiple complementary enterprises. Coconut palms, serving as the main crop, create a favorable microclimate that enables successful intercropping with a diverse range of crops including annuals and perennials, with the different canopy layers effectively utilizing available sunlight, nutrients, and soil moisture at varying depths.

Within this system, agroforestry components such as timber trees *Tectona grandis* (Teak), *Swietenia macrophylla* (Mahogany), and *Acacia mangium* (Acacia) are strategically integrated along boundaries and interspaces, providing additional income through timber production while contributing to soil conservation, carbon sequestration, and biodiversity enhancement. These trees also serve as windbreaks and sources of fuelwood for household energy needs.

The system incorporates well-planned pasture areas beneath coconut palms where various grass species and leguminous fodder crops are grown, supporting livestock integration through cattle, goats, or sheep rearing, which provides additional income through meat and milk production and contributes organic manure to enhance soil fertility. Beekeeping can seamlessly be integrated into this ecosystem, with coconut inflorescence and various intercrops providing excellent nectar sources throughout the year, resulting in honey production and enhanced crop pollination that boosts overall system productivity.

The integration of livestock enables the establishment of biogas units that convert animal waste into clean cooking fuel and high-quality organic manure, creating a circular economy within the farm. This integrated approach offers numerous benefits including enhanced land productivity, improved soil health through organic matter addition and nutrient cycling, increased biodiversity, better resource use efficiency, reduced input costs through synergistic interactions, enhanced carbon sequestration and climate resilience, improved income stability through product diversification, regular cash flow from different enterprises, reduced risk from crop failure or market fluctuations, enhanced food and nutritional security for farming families, better employment generation throughout the year, and improved environmental sustainability through reduced chemical inputs and enhanced ecosystem services.

The livestock component provides manure, helps in weed control, and utilizes crop residues, while the biogas unit reduces dependence on external energy sources and provides clean cooking fuel, with the slurry serving as an excellent organic manure that improves soil health and reduces the need for chemical fertilizers. The multilayer canopy structure of this integrated system also helps in better moisture conservation, reduced soil erosion, improved soil organic matter content, and enhanced biodiversity including beneficial insects and natural predators that contribute to natural pest control, making it an exemplary model of sustainable agriculture that optimizes economic returns while ensuring environmental conservation and social benefits, particularly suitable for small and marginal farmers in tropical and subtropical regions where coconut cultivation is prevalent.

Agroforestry in Coconut Plantation with Timber and Fuelwood

Coconut-based agroforestry systems integrating timber and fuelwood species represent an optimal approach to sustainable land use, maximizing economic returns and ecological benefits. In this system, coconut palms serve as the primary crop, providing continuous harvest while forming the upper canopy layer. High-value timber species such as *Tectona grandis* (Teak), *Swietenia macrophylla* (Mahogany), and *Acacia mangium* (Acacia) can be strategically integrated along boundaries and interspaces, chosen for their moderate shade tolerance and significant economic potential through quality timber production.

Fast-growing fuelwood species such as *Gliricidia sepium* (Gliricidia) and *Leucaena leucocephala* (Ipil-ipil) complement the system, offering quick returns through sustainable bioenergy production while enhancing soil fertility through nitrogen fixation. This diversified approach significantly reduces the risks associated with monoculture while generating multiple income streams through various products. The system delivers substantial environmental benefits, including improved soil health through leaf litter deposition and nitrogen fixation, enhanced biodiversity by creating habitats for various species, and climate benefits through increased carbon sequestration and microclimate regulation.



Gliricidia grown under Coconut

Success in implementing this system depends on careful species selection based on local conditions and market demand, optimal spacing considerations to minimize competition, and proper management practices including regular monitoring and maintenance. Through this integrated approach, coconut farmers can effectively balance economic prosperity with environmental conservation, creating a resilient and sustainable agricultural system that serves both immediate and long-term objectives.

Bee Keeping for Enhanced Pollination and Additional Income

Beekeeping in coconut plantations creates a mutually beneficial relationship that enhances honey production and coconut yield. Bees utilize coconut palm inflorescence as a primary nectar source, producing high-quality honey while simultaneously improving coconut pollination through their foraging activities. The partial shade provided by coconut palms creates ideal conditions for bee colonies, protecting them from extreme temperatures and other harsh weather conditions.

Beekeepers typically place hives strategically at the base of coconut palms or on elevated platforms to maximize nectar collection. The practice requires minimal space and complements coconut farming without competing for resources.

Beekeepers can harvest multiple products, including honey, beeswax, propolis, and pollen, creating diverse income streams. The increased



Bee keeping in coconut plantation

pollination from bees can enhance coconut yield significantly, with studies showing improvements up to 40%. Modern beekeeping techniques employ movable-frame hives that allow for easy honey extraction and colony management. Beekeepers must maintain proper hygiene standards, regularly monitor pests and diseases and ensure that adequate flowering plants are available year-round. Additional flowering plants can be introduced between coconut rows to provide supplementary nectar sources during off-seasons. The practice also supports broader ecosystem health by maintaining pollinator populations for other crops and natural vegetation. Regular maintenance includes cleaning hives, replacing old combs, controlling varroa mites, and ensuring adequate water supply near the colonies. The economic returns from beekeeping can significantly enhance farm income while contributing to biodiversity conservation.

Vermicompost for Improving Soil Fertility

Vermicomposting under coconut cultivation is a practical and sustainable approach to waste management and soil fertility enhancement. It utilizes fallen fronds, husks, weed slashes, and animal manure to create high-quality organic fertilizer through the biological activity of specific earthworm species, notably *Eisenia fetida*.

Coconut plantations naturally offer a shaded and moist environment ideal for earthworm survival and reproduction. This makes them well-suited for establishing on-site vermicomposting systems that require minimal input but deliver significant ecological and agronomic benefits.

Production Procedure for Vermicomposting

1. Site Selection and Infrastructure:

Choose a shaded location within the coconut plantation. Construct cement or brick-lined tanks with dimensions approximately 1.8 m wide and 0.9 m deep. A concrete base is essential to prevent nutrient leaching and worm escape.

2. Raw Material Preparation:

Collect and pre-compost organic waste for 10–15 days to initiate microbial activity. Suitable materials include coconut fronds, shredded husks, cover crop residues, and cow dung. The ideal feed should have a balanced C:N ratio (20:1), moderate moisture, and near neutral pH.

3. Layering:

In the prepared tank, place a 15 cm layer of coarse plant material (weeds or husks) to aid aeration. Follow this with alternate layers of cow dung and plant waste up to the desired height.

4. Inoculation with Earthworms:

Introduce *Eisenia fetida* at a recommended stocking rate of 2.5–5 kg/m². These species are preferred for their rapid reproduction and high organic matter conversion efficiency.

5. Moisture and Temperature Management:

Maintain 60–75% moisture and temperatures between 20–30°C. Regularly sprinkle water and provide ventilation. Avoid waterlogging and extreme heat.

6. Feeding and Turning (if necessary):

Earthworms feed continuously; ensure a steady supply of organic matter. While turning is optional, minimal disturbance ensures optimal worm activity. Avoid fresh, high-nitrogen waste that may heat up the bed.

7. Harvesting the Vermicompost:

After 2–5 months, the compost is ready. Shift fresh feed to one end of the tank to encourage worm migration. Harvest the dark, granular vermicompost from the vacated side. This nutrient-rich product contains N, P, K, micronutrients, enzymes, and beneficial microbes.

8. Continuous Production Cycle:

Replenish the raw materials and repeat the process. Worm populations multiply rapidly under ideal conditions, enabling large-scale and ongoing compost generation.

Benefits of Coconut Cultivation

1. Soil Enrichment: Vermicompost improves soil texture, aeration, and water retention.
2. Enhanced Fertility: Supplies essential nutrients (N, P, K, Ca, Mg) and beneficial microbes.
3. Pest and Disease Resistance: Promotes healthy root development and natural plant defences.
4. Economic Viability: Reduces fertilizer costs and creates potential for income through vermiwash and worm sales.
5. Environmental Sustainability: Reduces waste, sequesters carbon, and recycles nutrients within the plantation ecosystem.

Boosting Natural Predators for Biodiversity



Vermicomposting with *Eisenia fetida*

Natural predators play a vital role in maintaining healthy coconut ecosystems by controlling pest populations and contributing to overall biodiversity. Implementing strategies to increase these beneficial organisms can lead to more sustainable and productive coconut plantations while reducing the need for chemical pesticides.

Key natural predators in coconut ecosystems include birds like owls and woodpeckers, which control rodent populations that damage young coconuts and seedlings. Predatory insects such as ladybirds and praying mantises help manage destructive pests and scale insects. Spiders and predatory mites also contribute significantly to pest control.

To enhance natural predator populations, farmers can implement several practices: Creating habitat diversity by maintaining undergrowth vegetation and incorporating companion plants, providing shelter and alternative food sources for predatory insects and birds. Installing nesting boxes and perches encourages birds to establish permanent residence in the plantation. Establishing hedge rows and maintaining native vegetation strips between coconut rows creates ecological corridors that allow natural predators to move freely throughout the plantation. This practice also supports other beneficial organisms like pollinators.

Minimizing or eliminating chemical pesticides is crucial because they often harm beneficial predators more than the target pests. Instead, adopting Integrated Pest Management (IPM) approaches allows natural predator populations to thrive and establish sustainable pest control. Adding organic mulch and maintaining soil health supports ground-dwelling predators like beetles and spiders. These practices enhance natural pest control and contribute to broader ecosystem services. The increased biodiversity helps to create a more resilient agricultural system, improves soil health, enhances pollination, and provides additional environmental benefits while maintaining or improving coconut production.

Energizing through Biogas

Integrating biogas production within coconut plantations with integrated livestock is a sustainable, efficient approach that leverages natural synergies. "Coconut plantations provide an ideal setting for livestock, particularly cattle, goats, and poultry, as these animals can graze on the undergrowth vegetation and benefit from the shade provided by the coconut canopy, while simultaneously contributing to weed control and natural fertilization of the soil through their manure deposits.

This integrated system enables biogas production, as the animal waste can be collected and processed in anaerobic digesters, which break down the organic matter through microbial action in the absence of oxygen to produce biogas, a mixture primarily composed of methane (50-70%) and carbon dioxide (30-50%). The biogas production process in this integrated system begins with the collection of animal manure, which is mixed with water to create a slurry with the optimal moisture content for anaerobic digestion, typically 8-10% total solids.

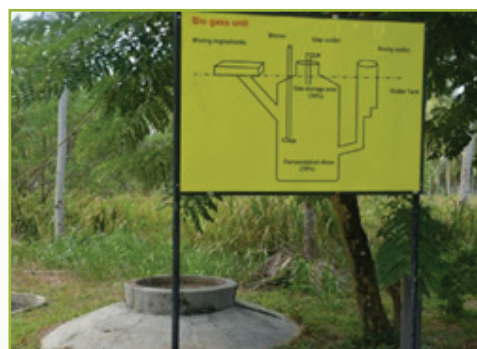
The anaerobic digestion process occurs in stages, beginning with hydrolysis, where complex organic compounds are broken down into simpler molecules, followed by acidogenesis and acetogenesis, where these molecules are further converted into acetic acid, hydrogen, and carbon dioxide, and finally methanogenesis, where methane-producing bacteria convert these compounds into biogas. The resulting biogas can be utilized directly for cooking, lighting, or electricity generation through biogas-powered generators, providing a renewable energy source for the farm operations and nearby communities. Furthermore, the digested slurry that remains after biogas production, known as digestate, serves as an excellent organic manure that can be applied back to the coconut plantation, creating a closed-loop nutrient cycling system that reduces the need for synthetic fertilizers and improves soil fertility.

This integrated approach also provides economic benefits through reduced energy costs, potential income from surplus biogas or electricity sales, and improved coconut and livestock productivity due to better soil management and organic fertilization.

The system's efficiency can be further enhanced by implementing proper management practices, such as maintaining optimal temperature ranges (35-37°C for mesophilic digestion) within the digesters, ensuring proper pH levels, and maintaining appropriate hydraulic retention times to allow for complete digestion of organic matter.

The integration of these components in coconut cultivation, livestock raising, and biogas production creates a resilient and sustainable agricultural system that maximizes resource utilization while minimizing environmental impact through the production of renewable energy and the recycling of nutrients.

This integration model can be particularly beneficial in tropical and subtropical regions where coconut cultivation is prevalent, enhancing farm productivity, energy security, and environmental sustainability while providing additional income through diversified production and reduced input costs. The success of integrated systems depends on careful planning and management, including proper sizing of the biogas digesters based on the available feedstock, maintaining appropriate animal stocking rates to ensure sustainable grazing without damaging the productivity of the coconut plantation, and implementing effective collection and handling systems for both animal waste.



A model of a biogas unit

Coconut Harvesting in Sri Lanka

Optimal Timing for Coconut Harvesting

The coconut palm is a perennial crop with a long lifespan that provides continuous yields over many years. Once it reaches full bearing, it produces 12 to 14 bunches annually. Therefore, at least one bunch matures each month throughout the year. The growth cycle of a coconut bunch lasts for 38 months, from the initiation of the inflorescence primordium to the full maturity of the nuts.

- The pre-fertilization phase lasts for 27 months.
- Fertilization and post-fertilization phases start with spathe opening and last for 10-11 months, resulting in a mature bunch of coconuts.

Therefore, the correct stage of maturity for coconut harvesting is 10-11 months after the inflorescence opens. At this stage, the coconuts are fully mature, and their quality is optimal for various uses.

Characteristics of Mature Coconuts Ready for Harvest

Several characteristics help to identify a mature coconut.

- The outer husk is brown or yellowish-brown, harder and more fibrous compared to tender nuts.
- The amount of coconut water will be reduced with maturity. When tapped, they produce a hollow sound.
- The meat turns firm, white and oily, separating from the shell easily.

Frequency of Coconut Harvesting

In Sri Lanka, coconuts are harvested at different intervals depending on factors such as,

- Coconut variety
- Plantation management
- Climatic conditions

The most common harvesting schedules are:

- Monthly (30-day)
- Bimonthly (60-day)
- 45-day cycles

Each harvesting interval has distinct benefits and drawbacks depending on:

- Cost
- Farm management
- Labour availability
- Market demand

Studies have shown that there is an increase in yield when monthly harvesting is completed, due to the reduction of button nut fall and losses from theft of fallen nuts. However, growers are reluctant to adopt this method due to the scarcity of pickers and the additional cost incurred in the additional six picks. As a result, most Sri Lankan farmers traditionally follow a bimonthly harvesting schedule.

A key drawback of bimonthly harvesting is the retention of an extra bunch on the tree between harvest cycles. This mature bunch continues to compete for plant assimilates with younger developing bunches, potentially reducing overall productivity. Also, leaving the mature nuts on the palm for an extra month can lead to yield losses due to fallen nuts, germination, and theft. However, when considering the labour cost, bimonthly harvesting is more profitable.

At 45-day intervals, one bunch may be fully mature while another may not yet be ready for harvest. Therefore, harvesting needs to be carried out carefully to ensure only mature coconuts are picked. Although this method requires more attention to detail, it generally needs to incur lower labour costs compared to the monthly harvesting.

Farmers can therefore decide the harvesting frequency based on the

- Size of the land
- Method of picking
- Price per nut
- Other types of costs, such as transport and nut collection

Methods of Coconut Harvesting

In Sri Lanka, Coconut harvesting is primarily done manually due to the tall nature of coconut palms. The two most common methods of harvesting coconuts are,

- Pole picking
- Climbing method

In the climbing method, typically, climbers ascend from 9 to 24 m in height above the ground, using rope. This is a commonly used method for coconut harvesting. Alternatively, pickers use long bamboo poles with an attached knife, especially for palms that are not much tall. This method is faster, efficient, and less dangerous compared to climbing.

In recent years, various innovations have been introduced to enhance harvesting efficiency and ensure the safety of pickers. Climbing devices are one of them. These devices have been developed to assist harvesters in climbing palms more easily and safely. These consist of a harness and a set of foot and hand grips that allow the user to climb the tree with minimal effort.

In some countries, trained monkeys are used for harvesting coconuts to significantly reduce the risk to human harvesters. Also, coconut-cultivating countries are facing a major problem due to a shortage of skilled workers for coconut harvesting.

As a solution, some developed countries have used different types of intelligent machines, like drone and robotic technologies. One example is an autonomous coconut tree-climbing robot, which is easy to operate and portable for use in large plantations.

Nut Falling

Nut falling in the coconut occurs naturally at maturity or abnormally due to various physiological, nutritional, or environmental stresses.

Premature nut fall refers to the dropping of nuts before they reach maturity, typically 2–7 months after fertilization. This is a serious issue that affects coconut yield and economic viability.

Causes of Premature Nut Falling:

- **Nutrient Deficiencies:**
 - Potassium (K) deficiency
- **Moisture Stress:**
 - Drought conditions or irregular irrigation, especially in sandy or shallow soils
- **Pollination and Fertilization Failure:**
 - Inadequate fertilization due to poor pollen viability, excessive rain, or heat stress
- **Pest Infestation and Diseases**
- **Hormonal Imbalance:**
 - Stress-induced hormonal changes, particularly in ethylene and auxin levels

Management and Mitigation

- **Balanced Fertilization:**
 - Apply adequate amounts of potassium and boron
- **Water Management:**
 - Maintain regular irrigation, especially during dry seasons.
 - Use mulching and cover crops to conserve soil moisture.
- **Pest and Disease Control:**
 - Regular monitoring and integrated plant health management strategies.
- **Good Agronomic Practices:**
 - Timely harvesting to avoid loss of mature nuts.
 - Avoid mechanical injuries during harvesting and pruning.

Annexure 1: Monocrop Coconut Cultivation

Notes:

- a) All spacing details in this annexure are based on the guidelines provided in the “Maximizing Profit through Intercropping, Animal husbandry and Integrated farming” section.
- b) The increase in coconut yield in intercropped coconut lands has not been taken into consideration.
- c) Intercrop cash flows include the costs and benefits of coconuts as well.
- d) md indicates man-days
- e) Errors in the figures due to rounding.

Benefit-cost of monocrop coconut cultivation (Rs/ac)

Assumptions:

Mature coconut plantation

Adopting the standard level of cultural practices

Annual nut production per palm	60
Number of palms per acre	64
Annual coconut yield	3,840

Activities	Unit	Unit Price (Rs)	Total (Rs)
INPUTS			
Fertilizer Application			
Adult Palm Mixture (3.3 kg per palm)	211.20	190.00	40,128.00
Dolomite (1 kg per palm)	64.00	26.00	1,664.00
Fertilizer Transport	1.00	750.00	750.00
Removing old mulch @ 40 palms/md	1.60	2,500.00	4,000.00
Broadcasting fertilizer @ 140 palms/md	0.50	2,500.00	1,143.00
Incorporating fertilizer into soil 35 palms/md	1.80	2,500.00	4,571.00
Replacing mulch @ 50 palms/md	1.30	2,500.00	3,200.00
Subtotal fertilizer application			55,456.00
HARVESTING CROP			
Harvesting by poles @ Rs 1200/pick/ac and 8 picks / yr	8.00	1,200.00	9,600.00
Collecting of nuts @ 500 nuts/man hour	0.96	2,500.00	2,400.00
Transporting to main heap Rs 1250/1000 nuts	3,840.00	1.00	4,800.00
Counting of nuts @ 8000 nuts/md	0.48	2,500.00	1,200.00
Stockpiling of nuts @1500/man hour	0.32	2,500.00	800.00
Cost for loading of nuts to tractor @500 nuts/man hour	0.96	2,500.00	2,400.00
Subtotal harvesting cost			21,200.00

Activities	Unit	Unit Price (Rs)	Total (Rs)
CULTURAL PRACTICES			
Weeding @ 15 coconut sq/md, 2 rounds per year	6.50	2,500.00	16,333.00
Husk Pits 1/5th each year (32 pits/ac)			
Excavation of pits	7.00	500.00	3,500.00
Husks for pits (500 husks * 7)	3,500.00	5.00	17,500.00
Covering up	7.00	750.00	5,250.00
Maintenance of fence- labour	5.00	2,500.00	12,500.00
Pest and disease control			9,132.00
Maintenance of fence- materials	1.00	5,000.00	5,000.00
Subtotal cultural practices			69,216.00
OTHER COSTS			
Management and Supervision	1.00	20,000.00	20,000.00
Road Office and Residence maintenance	1.00	1,500.00	1,500.00
Electricity	1.00	4,000.00	4,000.00
Taxes, Building upkeep, and tools	1.00	5,000.00	5,000.00
Administration	1.00	10,000.00	10,000.00
Subtotal other costs			40,500.00
Total Cost			186,372.00
Cost per nut			48.53
OUTPUTS			
Coconuts	3,840.00	80.62	318,720.00
Gross Income	318,720.00		
Net income	132,348.00		
Benefit-Cost Ratio	1.71		

Annexure 2: Ginger Cultivation Under Coconut

Gross margin (Rs/ac) of Ginger (Variety: Rangoon)

Activities	Year 1
OUTPUTS	
Ginger (kg/ac)	2,340.00
Average price (Rs/kg)	500.00
Income from ginger	1,170,000.00
Coconut production (nuts/ac)	3,840.00
Average price (Rs/nut)	80.62
Income from coconuts (Rs/ac)	309,580.00
Total income	1,479,580.00
INPUTS	
Labour cost (Rs/ac)	
Land clearing, planting and bed preparation	7.00
Preparation of seeds, planting and application of basal fertilizer mixture	4.00
Collecting coconut fronds for mulching	11.00
Cleaning drains and compacting the soil around the plants (2times)	12.00
Weeding (3times)	38.00
Application of chemical fertilizer (2times)	3.00
Application of fungicide and pesticide	1.00
Harvesting and processing	25.00
Other	7.00
Total labour use (md/ac)	108.00
Wage rate (Rs/md)	2,500.00
Subtotal labor cost (Rs/ac)	270,000.00

Material cost (Rs/ac)	Year 1
Planting material (kg)	390.00
Price of planting material (Rs/kg)	800.00
Cost of planting materials	312,000
Inorganic fertilizers	10,171
Cost of Insecticides and Fungicide per bed	150
Number of beds	75
Cost of Insecticides and fungicides	11,250
Subtotal cost for materials	333,421
Transport cost (Rs/ac)	Year 1
Transport of planting material	1,500
Fertilizer transport	1,500
Subtotal transport cost	3,000
Machinery cost (Rs/ac)	
Land preparation (Rs/ac) - disk plough	5,043
Land preparation (Rs/ac) - tine tiller	3,104
Subtotal machinery cost	8,147
Total cost of ginger cultivation (Rs/ac)	614,569
Cost of coconuts	186,372
Total cost (including coconut)	800,941
Gross Margin (Rs/ac)	678,640
Benefit-Cost ratio	1.85

Annexure 3: Turmeric Cultivation Under Coconut

Gross Margin (Rs/ac) of Turmeric

Activities	Year 1
OUTPUTS	
Turmeric (kg/ac)	3,120
Average price (Rs/kg)	320
Income from turmeric (Rs/ac)	998,400
Yield of coconuts (Nuts/ac)	3,840
Average price (Rs/nut)	80.62
Income from coconuts (Rs/ac)	309,580
Total income (Rs/ac)	1,307,981
INPUTS	
Labour cost (Rs/ac)	
Land clearing and planting bed preparation	7
Preparation of seeds, planting and application of basal fertilizer	7
Cleaning drains and compacting the soi around the plants (2times)	12
Collecting coconut fronds and mulching	6
Fertilizer application	2
Weeding (3times)	32
Application of fungicides and pesticides	1
Harvesting and processing	19
Other	7
Total labour use (md/ac)	93
Wage rate (Rs/md)	2,500
Subtotal labour cost	232,500.00

Material cost (Rs/ac)	Year 1
Planting material (kg)	312
Price of planting material (Rs/kg)	400
Cost of planting material (Rs)	124,800
Cost of fertilizer (Rs)	13,264
Pesticides and fungicides (Rs)	11,250
Subtotal material cost	149,314
Transport cost (Rs/ac)	
Transport of planting material	1,000.00
Fertilizer transport	500.00
Subtotal transport cost	1,500.00
Machinery Cost (Rs/ac)	Year 1
Land preparation (Rs/ac) - disk plough	2,522.00
Land preparation (Rs/ac) - tine tiller	1,552.00
Subtotal machinery cost	4,074.00
Total cost (Rs/ac)	386,579.00
Cost of coconuts	186,372.00
Total costs (including coconut)	5,729,511.00
Gross Margin (Rs/ac)	735,030.00
Benefit-Costratio	2.3

Annexure 4: Cassava Cultivation Under Coconut

Gross Margin (Rs/ac) of Cassava (Variety: Kirikawadi)

Activities	Year 1
OUTPUTS	
Cassava (kg/ac) Average 3 kg per plant	3,948.00
Average price (Rs/kg)	80.00
Income from Cassava (per 6 months)	315,840.00
Yield of coconuts (nuts/ac)	3,840.00
Price of coconuts (Rs/nut)	80.62
Income from coconuts (Rs/ac)	309,580.00
Total income (including coconuts)	625,420.00
INPUTS	
Labour cost (Rs/ac)	
Preparation of Land	1.00
Planting	2.00
Top-dressing fertilizer application	1.00
Weeding and earthing up (2 times)	26.00
Pest and disease control	1.00
Harvesting and processing	17.00
Total labour use (md/ac)	48.00
Wage rate (Rs/md)	2,500.00
Subtotal labour cost	120,000.00

Material cost (Rs/ac)	Year 1
Cost for cuttings (1316 cuttings @ Rs 8/ cutting)	10,528.00
Cost of inorganic fertilizers	19,203.00
Cost of plant protection materials	3,509.00
Subtotal material cost	33,240.00
Transport cost (Rs/ac)	
Transport cost of planting material	2,000.00
Fertilizer transport cost	1,000.00
Subtotal transport cost	3,000.00
Machinery cost	
Land preparation with 4WT disc plough (Rs/ac)	3,381.00
Tilling	2,080.00
Subtotal	5,461.00
Totalcost (Rs/ac per 6 months)	161,701.00
Cost of coconuts	186,372.00
Total cost (including coconuts) (Rs/ac)	277,347.00
Gross Margin (RS/AC)	277,347.00
Benefit-Cost Ratio	2.25

Annexure 5: Papaya Cultivation Under Coconut

Gross Margin (Rs/ac) of Papaya (Variety: Red Lady)

Activities	Years		
OUTPUTS	1	2	3
Fruits (kg/ac)	12,768.00	21,280.00	8,512.00
Average price (Rs/Kg)	80.00	80.00	80.00
Income from papaya (Rs/ac)	1,021,440.00	1,702,400.00	680,960.00
Yield of Coconuts (Nuts/ac)	3,840.00	3,840.00	3,840.00
Price of coconuts (Rs/nut)	80.62	80.62	80.62
Income from coconuts (Rs/ac)	309,580.00	309,580.00	309,580.00
Total income (Rs/ac)	1,331,020.00	201,980.00	990,540.00
INPUTS			
Labour cost (Rs/ac)			
Digging holes	12.00		
Preparation and planting	11.00		
Mulching	11.00	10.00	10.00
Fertilizer application	15.00	15.00	15.00
Weeding (2 times per year)	5.00	5.00	5.00
Irrigation	42.00	42.00	42.00
Pest and disease control	5.00	5.00	5.00
Harvesting	85.00	142.00	58.00
Total labour use (md/ac)	186.00	220.00	135.00
Wage rate	2,500.00	2,500.00	2,500.00
Subtotal labor cost (Rs/ac)	465.00	547,500.00	337,500.00
Materials cost (Rs/ac)			
Number of plants	266.00		
Average price (Rs/plant)	50.00		
Cost of plants	13,300.00		
Cost of fertilizer	265,016.00	189,020.00	189,020.00

Activities	Years		
INPUTS	1	2	3
Transport cost (Rs/ac)			
Cost of irrigation-hose irrigation line	80,000.00		
Cost of pesticides and fungicides	11,970.00	11,970.00	11,970.00
Subtotal materials	370,286.00	200,990.00	200,990.00
OUTPUTS			
Transport cost (Rs/ac)			
Planting materials	1,500.00		
Fertilizer	1,500.00	1,500.00	1,500.00
Sub-total transport cost	3,000.00	1,500.00	1,500.00
Machinery cost (Rs/ac)			
Cost for slashing	4,000.00		
Total variable cost	842,286.00	749,990.00	539,990.00
Cost for coconuts	186,372.00	186,372.00	186,372.00
Cost including coconuts	1,028,657.00	936,362.00	726,362.00
Gross Margin (Rs/ac)	302,363.00	1,075,619.00	264,179.00

At 10% discount rate

Net Present Value (Rs./ac) 1,498,528.00

Benefit-Cost ratio 1.6

Annexure 6: Black Pepper Cultivation Under Coconut

Gross margin (Rs/ac) of Pepper

Activities	Years									
OUTPUTS	1	2	3	4	5	6	7	8 to 19	20 to 40	
Expected average yield (kgdry/ vine/year)			0.05	0.30	0.50	0.60	0.80	1.00	1.50	
No. of good bearing vines (max.90%ofplantingstand)			255.00	332.00	391.00	391.00	391.00	391.00	260.00	
Average yield (GR1drykg/ac)			13.00	100.00	195.00	234.00	312.00	391.00	391.00	
Price-GR (Rs1850/kg)										
Gross income			23,615.00	184,195.00	361,305.00	433,566.00	578,088.00	722,610.00	722,610.00	
Yield of coconuts (nuts/ac)	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	
Average price (Rs/nut)	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	
Income from coconuts (Rs/ac)	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	
Total income	309,580.00	309,580.00	333,195.00	493,775.00	670,885.00	743,146.00	887,668.00	1,032,190.00	1,032,190.00	
INPUTS										
Labour cost (Rs/ac)										
Cutting pits	18.00									
Planting of gliricidia	9.00									
Preparation and planting	26.00									
Mulching and temporary shading	9.00									

Activities		Years									
INPUTS		1	2	3	4	5	6	7	8 to 19	20 to 40	
Filling vacancies, mulching, and shading		2.00	2.00								
Weeding (3 times a year, by grass cutter)		33.00	33.00	33.00	33.00	33.00	33.00	33.00	33.00	33.00	
Gliricidia pruning and application of lopping		18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	
Training pepper vines		9.00	9.00	9.00							
Fertilizer application		5.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00	10.00	
Pruning vines and plant sanitation				4.00	4.00	7.00	9.00	9.00	9.00	9.00	
Pest and disease control		4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	
Harvesting and threshing				2.00	12.00	23.00	28.00	37.00	47.00	47.00	
Drying				1.00	3.00	7.00	9.00	12.00	15.00	15.00	
Other activities		3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	
Total labour units		136.00	80.00	85.00	88.00	106.00	115.00	127.00	140.00	140.00	
Wage rate (per day)		2,500.00									
Subtotal		340,000.00	200,000.00	212,500.00	220,000.00	265.00	287,500.00	317,500.00	350,000.00	350,781.00	
Material Cost (Rs)											
Number of plants		434.00									
Cost per plant		30.00									

Activities	Years									
INPUTS	1	2	3	4	5	6	7	8 to 19	20 to 40	
Cost of plants	13,020.00									
Price of Gliricidia poles	30.00									
Cost of Gliricidia poles	13,020.00									
Pesticides and fungicides	10,850.00	10,850.00	10,850.00	10,850.00	10,850.00	21,700.00	21,700.00	21,700.00	21,700.00	
Cost of fertilizer	75,828.00	20,736.00	29,030.00	29,030.00	29,030.00	29,030.00	29,030.00	29,030.00	29,030.00	
Subtotal	112,718.00	31,586.00	39,880.00	9,880.00	39,880.00	50,730.00	50,730.00	50,730.00	50,730.00	
Machinery Cost (Rs)										
Slashing	2,000.00									
Fertilizer transport cost	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	
Transport cost of planting materials	5,000.00									
Subtotal	8,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	
Total variable cost (Rs/ac)	460,718.00	232,586.00	253,380.00	260,880.00	305,880.00	339,230.00	369,230.00	401,730.00	402,511.00	
OUTPUTS	1	2	3	4	5	6	7	8 to 19	20 to 40	
Cost of coconuts	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	
Total cost (including coconuts)	647,090.00	418,958.00	439,752.00	447,252.00	492,252.00	525,602.00	555,602.00	588,102.00	588,883.00	
Gross Margin (Rs/ac)	-337,509.00	-109,377.00	-106,556.00	46,524.00	178,634.00	217,545.00	332,067.00	444,089.00	443,308.00	

At 10% discount rate

Net Present Value (Rs./ac) 11,876,943.00

Benefit-Cost ratio 1.6

Internal rate of return (%) = 31

Annexure 7: Pineapple Cultivation Under Coconut

Gross Margin (Rs/ac) of Pineapple

Activities	Years				
OUTPUTS	1	2	3	4	5
Yield (kg)		4,441.00	4,441.00	6,832.00	5,466.00
Average price (Rs/kg)		350.00	350.00	270.00	200.00
Income from fruits (Rs/ac)		1,554,280.00	1,554,280.00	1,844,640.00	1,093,120.00
Number of suckers		6,832.00	6,832.00	6,832.00	6,832.00
Average price (Rs/sucker)		45.00	45.00	45.00	45.00
Income from suckers (Rs/ac)		307,440.00	307,440.00	307,440.00	307,440.00
Income from pineapple		1,861,720.00	1,861,720.00	2,152,080.00	1,400,560.00
Yield of coconuts (Nuts/ac)	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00
Average price (Rs/nut)	80.62	80.62	80.62	80.62	80.62
Income from coconuts (Rs/ac)	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00
Total income (including coconuts)	309,580.00	2,171,300.00	2,171,300.00	2,461,660.00	1,710,140.00
Labour use (md/ac)					
Land clearing	2.00				
Dipping of suckers in chemicals	2.00				
Planting	6.00				
Fertilizer application and adding soil	27.00	27.00	27.00	27.00	
Earthing up and manual weeding (2 times per year)	34.00	34.00	34.00	34.00	
Application of insecticides and weedicides	3.00	3.00	3.00	3.00	
Application of hormones	2.00	4.00	4.00	4.00	
Harvesting		10.00	20.00	20.00	
Removal of suckers		11.00	11.00	11.00	
Removal of leaves		4.00	4.00	4.00	
Total labour use (md)	76.00	93.00	103.00	103.00	
Wage rate (Rs/md)	2,500.00				
Subtotal labour cost (Rs/ac)	190,000.00	232,500.00	257,500.00	257,500.00	87,500.00

Activities	Years				
OUTPUTS	1	2	3	4	5
Material cost (Rs/ac)					
Number of suckers	3,416.00				
Average price (Rs/sucker)	45.00				
Cost for suckers	153,720.00				
Cost for fertilizer	112,728.00	88,133.00	88,133.00	88,133.00	
Cost of pesticides	8,540.00	8,540.00	17,080.00	17,080.00	17,080.00
Cost for Ethrel	1,281.00	1,281.00	1,281.00	2,562.00	2,562.00
Cost of weedicides	25,620.00	25,620.00	25,620.00	25,620.00	25,620.00
Subtotal material cost	301,889.00	123,574.00	132,114.00	133,395.00	45,262.00
Transport cost (Rs/ac)					
Transport cost of planting material	6,000.00				
Fertilizer transport cost	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00
Subtotal transport cost	7,000.00	1,000.00	1,000.00	1,000.00	1,000.00
Machinery cost (Rs/ac)					
Cost of ploughing	6,500.00				
Cost of harrowing	4,000.00				
Sub total	10,500.00				
Total variable cost (Rs/ac)	508,389.00	357,074.00	390,614.00	391,895.00	133,762.00
Cost of coconuts	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00
Total cost (including coconuts)	694,705.00	542,713.00	576,873.00	578,154.00	319,742.00
Gross Margin (Rs/ac)	-386,180.00	1,627,855.00	1,594,315.00	1,883,394.00	1,390,007.00

At 10% discount rate

Net Present Value (Rs./ac) 3,341,228.00

Benefit-Cost ratio 2.4

Internal rate of return (%) = 5.2

Annexure 8: Banana Cultivation Under Coconut

Gross Margin (Rs/ac) of Banana

Activities	Years				
OUTPUTS	1	2	3	4	5
Yield of Ambul (kg/ac)		1,553.00	2,330.00	1,747.00	1,035.00
Yield of Seeni (kg/ac)		1,206.00	1,535.00	1,450.00	1,157.00
Yield of Kolikuttu (kg/ac)		1,980.00	2,475.00	2,475.00	1,716.00
Average price of Ambul (Rs/kg)	100.00				
Average price of Seeni (Rs/kg)	120.00				
Average price of Kolikuttu (Rs/kg)	80.00				
Income from Banana		854,438.00	1,110,166.00	1,041,638.00	722,898.00
Suckers (Ambul)		134.00	134.00	134.00	134.00
Suckers (Seeni)		134.00	134.00	134.00	134.00
Suckers (Kolikuttu)		300.00	300.00	300.00	300.00
Income from suckers (Rs/ac)		61,080.00	61,080.00	61,080.00	61,080.00
Gross return from bananas		915,518.00	1,171,246.00	1,102,718.00	783,978.00
Coconut production (nuts/ac)		3,840.00	3,840.00	3,840.00	3,840.00
Price of coconuts (Rs/nut)		80.62	80.62	80.62	80.62
Income from coconuts	309,580.00	309,580.00	309,580.00	309,580.00	390,580.00
Total income	309,580.00	1,225,098.00	1,480,827.00	1,412,298.00	1,093,559.00
INPUTS					
Labour Cost (md/ac)					
Digging holes	12.00				
Fertilizer application per year	15.00	13.00	13.00	13.00	13.00
Dipping of suckers in chemicals	1.00				
Preparation and planting	11.00				
Removal of suckers		6.00	6.00	6.00	6.00
Weeding (manual, 2 times per year)	2.00	2.00	2.00	2.00	2.00
Application of insecticides and pesticides	3.00	3.00	3.00	3.00	3.00

Activities	Years				
OUTPUTS	1	2	3	4	5
Removal of old banana stems and leaves	5.00	27.00	27.00	27.00	27.00
Harvesting		7.00	7.00	7.00	10.00
Other activities	3.00	3.00	3.00	3.00	3.00
Total labour use (md/ac)	52.00	61.00	61.00	61.00	64.00
Wage rate	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00
Subtotal	128,138.00	152,500.00	152,500.00	152,500.00	160,000.00
Material cost (Rs/ac)					
Number of suckers (Kolikuttu, spacing)	132.00				
Number of suckers (Ambul, spacing)	67.00				
Number of suckers (Seeni, spacing)	67.00				
Average price (Rs/sucker) (Kolikuttu)	150.00				
Average price (Rs/sucker) (Seeni+Ambul)	60.00				
Cost of suckers	27,840.00				
Cost of fertilizer	320,237.00	167,420.00	167,420.00	167,420.00	167,420.00
Cost of pesticides	6,650.00	3,990.00	390.00	3,990.00	3,990.00
Subtotal	354,727.00	171,410.00	171,410.00	171,410.00	171,410.00
Transportation cost (Rs/ac)					
transport cost of planting material	3,500.00				
Fertilizer transport cost	4,000.00	2,000.00	2,000.00	2,000.00	2,000.00
Subtotal	7,500.00	2,000.00	2,000.00	2,000.00	2,000.00
Machinery cost (Rs/ac)					
Slashing	6,500.00				
Subtotal	6,500.00				
Total cost (Rs/ac)	498,865.00	325,910.00	325,910.00	325,910.00	333,410.00

Activities	Years				
OUTPUTS	1	2	3	4	5
Cost of coconut	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00
Total cost (including coconuts)	683,237.00	12,282.00	12,282.00	12,282.00	19,782.00
Gross Margin (Rs/ac)	-373,657.00	712,817.00	968,545.00	900,016.00	573,777.00

At 10% discount rate

Net Present Value (Rs./ac) 1,637,605.00

Benefit-Cost ratio 2.07

Internal rate of return (%) = 20.6

Annexure 9: Anona Cultivation Under Coconut

Gross Margin (Rs/ac) of Anona

Activities	Year										
	1	2	3	4	5	6	7	8	9	10-15	16-20
OUTPUTS											
Yield of Anona (kg/ac)	-	787.58	1,260.00	2,362.58	3,937.58	5,315.65	5,315.65	5,315.65	5,315.65	5,512.50	4,567.52
Average price (Rs/kg)		90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00
Income from fruits	-	708.86	113,400.00	212,632.00	354,382.00	478,408.00	478,408.00	478,408.00	478,408.00	496,125.00	411,076.00
Yield of coconuts (Nuts/ac)	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00
Average price (Rs/nut)	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62
Income from coconuts (Rs/ac)	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00
Total income (including coconut)	309,580.00	380,463.00	422,980.00	522,213.00	663,963.00	787,988.00	787,988.00	787,988.00	787,988.00	805,705.00	720,657.00
INPUTS											
Labour cost (Rs/ac)											
Digging and planting	5.00										
Fertilizer application	7.00	9.00	12.00	12.00	15.00	15.00	15.00	15.00	15.00	15.00	15.00
Preparation and planting	7.00										
Weeding (manual, 2 times a year)	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Mulching (2 times per year)	3.00	3.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
Pest and disease control	2.00	2.00	3.00	4.00	4.00	5.00	5.00	5.00	5.00	7.00	9.00
Pollination		2.00	2.00	4.00	4.00	7.00	7.00	7.00	7.00	8.00	11.00
Training and pruning	2.00	2.00	4.00	5.00	5.00	7.00	7.00	7.00	7.00	7.00	7.00

Activities	Year										
	1	2	3	4	5	6	7	8	9	10-15	16-20
OUTPUTS											
Cost of slashing the grass	4,000.00										
Subtotal	4,000.00										
Total cost for Anona cultivation	166,053.00	139,933.00	186,061.00	234,257.00	288,770.00	364,600.00	364,484.00	367,110.00	367,110.00	379,725.00	369,725.00
Cost of coconut	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00
Total cost (including coconut)	352,425.00	326,305.00	372,433.00	420,629.00	475,142.00	550,972.00	550,856.00	553,481.00	563,482.00	556,097.00	556,097.00
Gross Margin (Rs/ac)	42,845.00	54,158.00	50,548.00	101,585.00	188,821.00	237,017.00	237,132.00	234,507.00	234,507.00	239,609.00	164,004.00

At 10% discount rate
Net Present Value (Rs./ac) 141,677.00
Benefit-Cost ratio 1.3
Internal rate of return (%) = 19

Annexure 10: Cashew Cultivation Under Coconut

Gross Margin (Rs/ac) of Cashew

Activities	Years										
	1	2	3	4	5	6-10	11-15	16-20	21-25		
OUTPUT											
Yield of cashew (kg/ac)	-	-	24.50	49.00	98.00	313.60	539.00	588.00	612.50		
Average price (Rs/kg)			600.00	600.00	600.00	600.00	600.00	600.00	600.00		
Yield of Coconuts (Nuts/ac)	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00		
Average price (Rs/nut)	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62		
Income from coconuts (Rs/ac)	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00		
Total income (including coconut)	309,580.00	309,580.00	324,280.00	338,980.00	368,380.00	497,740.00	632,980.00	662,380.00	677,080.00		
INPUTS											
Labour use (md)											
Digging holes	2.00										
Preparation and planting	5.00										
Fertilizing	1.00	2.00	2.00	2.00	3.00	3.00	3.00	3.00	3.00		
Weeding (manual)	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00		
Mulching	2.00	2.00	2.00	2.00							
Pest and disease control	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00		
Pruning		1.00	1.00	2.00	2.00	2.00	3.00	3.00	3.00		
Harvesting and drying		2.00	3.00	6.00	20.00	35.00	38.00	39.00			
Total labour use (md)	17.00	12.00	14.00	16.00	18.00	32.00	48.00	51.00	52.00		
Wage rate (Rs/md)	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00		
Subtotal labour cost	42,500.00	30.00	35,000.00	40,000.00	45,000.00	80,000.00	120,000.00	127,500.00	130,000.00		

Activities		Years									
INPUTS		1	2	3	4	5	6-10	11-15	16-20	21-25	
Materials cost (Rs/ac)											
Cost of plants	9,800.00										
Cost of fertilizer	13,286.00	2,429.00	3,481.00	3,731.00	5,570.00	8,271.00	12,407.00	16,542.00	20,678.00		
Fungicide and pesticide	3,267.00	3,267.00	4,900.00	4,900.00	4,900.00	6,533.00	9,800.00	13,067.00	16,333.00		
Subtotal material cost	26,353.00	5,695.00	8,381.00	8,631.00	10,470.00	14,805.00	22,207.00	29,609.00	37,011.00		
Transport cost (Rs/ac)											
Transport cost of planting materials (Rs/ac)(Rs/ac)	3,000.00										
Fertilizer transport cost	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	
Subtotal	7,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	
Machinery cost (Rs/ac)											
Cost for weeding	4,000.00										
Subtotal (Rs.ac)	4,000.00										
Total variable cost	79,853.00	39,695.00	47,381.00	52,630.00	59,470.00	98,805.00	99,503.00	146,207.00	161,109.00		
Cost of coconut	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	
Total cost (including coconut)	266,225.00	226,067.00	233,753.00	239,003.00	245,842.00	285,177.00	285,875.00	322,578.00	347,481.00		
Gross Margin (Rs/ac)	43,356.00	85,513.00	90,527.00	99,978.00	122,538.00	212,564.00	211,866.00	300,402.00	314,900.00		

At 10% discount rate

Net Present Value (Rs./ac) 1,827,004.00

Benefit-Cost ratio 1.66

Internal rate of return (%) = 29

Annexure 11: Mango Cultivation Under Coconut

Gross Margin (Rs/ac) of Mango

Activities	Years									
	1	2	3	4	5	6-9	10-15			
OUTPUT										
Yield of mango (kg/ac)										
Average price (Rs/kg)					7,350.00	8,779.00	19,192.00			
Income from fruits					100.00	100.00	100.00			
Yield of coconut (Nuts/ac)	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00			
Average price (Rs/nut)	80.62	80.62	80.62	80.62	80.62	80.62	80.62			
Income from coconuts (Rs/ac)	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00			
Total income (including coconut)	309,580.00	309,580.00	309,580.00	309,580.00	1,044,581.00	1,187,502.00	2,228,747.00			
INPUTS										
Labour use (md)										
Digging holes	2.00									
Preparation and planting	5.00									
Fertilizer application	6.00	7.00	9.00	9.00	10.00	7.00	7.00			
Weeding (manual, 3times per year)	6.00	6.00	6.00	6.00	6.00	6.00	6.00			
Mulching (2 times per year in first two years)	5.00	5.00								
Pest and disease control	2.00	2.00	3.00	3.00	5.00	5.00	5.00			
Pruning		2.00	3.00	4.00	4.00	4.00	4.00			
Harvesting					17.00	20.00	43.00			
Total labour use (md)	26.00	22.00	22.00	22.00	42.00	42.00	65.00			
Wage rate (Rs/md)	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00			

Activities	Years							
	1	2	3	4	5	6-9	10-15	
INPUTS								
Subtotal labour cost (Rs/ac)	65,000.00	55,000.00	55,000.00	55,000.00	105,000	105,000.00	162,500.00	
Materials cost (Rs/ac)								
Number of plants	49.00							
Average price per plant (graftedplant)	200.00							
Cost of plants	9,800.00							
Cost of fertilizer	23,741.00	19,825.00	23,353.00	26,470.00	298,51??	23,794.00	48,255.00	
Cost of fungicide and pesticide	4,900.00	4,900.00	7,350.00	7,350.00	14,700.00	14,700.00	14,700.00	
Total material cost	38,441.00	24,725.00	30,703.00	33,820.00	44,551.00	38,494.00	62,955.00	
Transport cost (Rs/ac)								
Transport cost of planting materials (Rs/ac)	2,000.00							
Fertilizer transport cost	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	
Subtotal transport cost	3,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	
Machinery cost								
Slashing of weeds	4,000.00							
Subtotal	4,000.00							
Cost for mango cultivation	110,941.00	81,225.00	87,203.00	90,320.00	151,050.00	162,289.00	226,955.00	
Cost ofr coconut (Rs/ac)	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	
Total cost (including coconut)	297,312.00	267,597.00	273,575.00	276,692.00	337,423.00	348,661.00	413,327.00	
Gross Margin (Rs/ac)	12,268.00	41,983.00	36,005.00	32,889.00	707,158.00	838,840.00	1,815,420.00	

At 10% discount rate

Net Present Value (Rs./ac) 6,109,542.00

Benefit-Cost ratio 3.2

Internal rate of return (%) = 71

Annexure 12: Cinnamon Cultivation Under Coconut

Gross Margin (Rs/ac) of Cinnamon

Activities		Years										
OUTPUTS	1	2	3	4	5	6	7	8	9	10	11to20	
Yield (dry/kg/bush/year)			0.04	0.05	0.08	0.11	0.13	0.13	0.13	0.13	0.13	
No. of yielding bushes (max.95%)			1,254.00	1,447.00	1,562.00	1,649.00	1,649.00	1,649.00	1,649.00	1,649.00	1,649.00	
planting standard			50.00	72.00	125.00	181.00	206.00	206.00	206.00	206.00	206.00	
Total dry yield			25.00	36.00	62.00	91.00	103.00	103.00	103.00	103.00	103.00	
Processing cost*			25.00	36.00	62.00	91.00	103.00	103.00	103.00	103.00	103.00	
Net yield for farmer												
Average price Rs.4,000												
Income from Cinnamon			100,302.00	144,667.00	249,984.00	362,824.00	412,300.00	412,300.00	412,300.00	412,300.00	4,123,300.00	
Income from Cinnamon sticks			12,538.00	14,467.00	15,624.00	16,492.00	16,492.00	16,492.00	16,492.00	16,492.00	16,492.00	
Yield of coconuts (nuts/ac)	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	3,840.00	
Price of coconut (Rs/nut)	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	80.62	
Income from coconut	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,580.00	309,680.00	309,580.00	309,580.00	
Total Income including coconut	309,580.00	309,580.00	422,420.00	468,714.00	575,189.00	688,897.00	738,373.00	738,373.00	738,373.00	738,373.00	738,373.00	
Labour use (md/ac)												
Cutting pits	35.00											
Filling holes and planting	29.00											

Activities		Years									
INPUTS	1	2	3	4	5	6	7	8	9	10	11to20
Aftercare and infilling	6.00	3.00	2.00	1.00							
Weeding	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00
Fertilizer application	21.00	42.00	42.00	42.00	42.00	42.00	42.00	42.00	42.00	42.00	42.00
Pruning and cleaning bushes		2.00	3.00	5.00	7.00	7.00	7.00	9.00	9.00	10.00	10.00
Plant protection	1.00	1.00	1.00	2.00	4.00	4.00	4.00	3.00	3.00	4.00	4.00
Other activities	1.00	1.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Total labour units	109.00	65.00	66.00	68.00	71.00	71.00	71.00	72.00	72.00	74.00	74.00
Wage rate	2,500.00										
Subtotal	272,500.00	162,500.00	165,000.00	170,000.00	177,500.00	177,500.00	177,500.00	180,000.00	180,000.00	185,000.00	185,000.00
Material cost											
Number of plants	1,736.00										
Cost per plant	18.00										
Total cost of plants	31,248.00										
Pesticide and fungicide	4,133.00	4,133.00	6,200.00	6,200.00	6,200.00	8,267.00	8,267.00	8,267.00	8,267.00	8,267.00	
Cost for fertilizer	5,653.00	9,744.00	14,616.00	14,616.00	14,616.00	14,616.00	14,616.00	14,616.00	14,616.00	14,616.00	
Subtotal	41,035.00	13,878.00	20,816.00	20,816.00	20,816.00	22,883.00	22,883.00	22,883.00	22,883.00	22,883.00	

Activities	Years										
	1	2	3	4	5	6	7	8	9	10	11to20
OUTPUTS											
Machinery cost(Rs/ac)											
Cost of Slashing the grass	2,000.00										
Fertilizer transportcost	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00
Planting materials transport cost	4,000.00										
Subtotal	7,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00
Total cost (Rs/ac)	321,034.00	177,878.00	187,316.00	192,316.00	199,816.00	201,882.00	201,882.00	204,383.00	204,383.00	209,383.00	186,500.00
Cost for coconut	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00	186,372.00
Total cost (including coconuts)	507,258.00	363,431.00	373,987.00	377,603.00	386,042.00	388,109.00	388,109.00	391,726.00	391,726.00	396,548.00	372,872.00
GROSS MARGIN (Rs./ac)	197,826.00	54,669.00	48,732.00	90,026.00	189,000.00	300,642.00	350,118.00	347,618.00	347,618.00	342,618.00	365,500.00

At 10% discount rate

Net Present Value (Rs./ac) 4,796,366.00

Benefit-Cost ratio 1.68

Internal rate of return (%) = 48

